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Review article

A survey and benchmark for household electricity forecasting: From statistical to foundation models

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ABSTRACT

Energy forecasting plays a crucial role in modern power systems, enabling accurate estimation of energy consumption and generation to support cost reduction, grid stability, and the integration of distributed energy resources such as photovoltaic systems, electric vehicles, and battery energy storage systems (BESS). While forecasting methodologies have evolved from traditional statistical and machine learning models to advanced transformer-based architectures, recent time series foundation models (TSFMs) introduce new capabilities such as zero-shot forecasting that have remained underexplored in current systematic benchmarks. In this paper, we address this gap by reviewing and benchmarking 30 state-of-the-art models across three residential energy forecasting tasks, i.e., load, photovoltaic generation, and BESS operation. Our study aims to (i) evaluate the performance of statistical, machine learning, deep learning models, and zero-shot TSFMs, (ii) identify the benefits, limitations, and trade-offs involved in selecting forecasting models for practical applications, and (iii) examine how model performance varies across different forecasting tasks. Using real-world datasets from Belgium, Germany, the Netherlands, and the United Kingdom, we apply a unified data pre-processing pipeline, default model configurations, and multiple temporal sampling rates, and evaluate the performance of these models using mean absolute error and root mean squared error. Our experimental results show that LightGBM provides the most reliable and accurate performance among trained models while Timer-XL demonstrate strong zero-shot forecasting capabilities among TSFMs. The source code is publicly available at https://github.com/dcdube/Forecast-Models-Evaluation.

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1. Introduction

Time series forecasting plays an important role in energy systems by enabling estimation of energy production and consumption, which in turn allows grid operators to balance supply and demand in real-time (Dong et al., 2025; Vivas et al., 2020). Given the high cost of energy storage systems, electricity is typically delivered to the consumers as soon as it is generated. Therefore, accurate forecasts with minimum error can lead to operational cost reduction and enhanced grid stability (Dong et al., 2025; Bot et al., 2021). Time series forecasting has a wide range of applications in power systems, including but not limited to load forecasting (Ahmad et al., 2022; Chandrasekaran, 2025; Habbak et al., 2023), renewable energy generation forecasting (Arslan Tuncar et al., 2024; Croonenbroeck and Stadtmann, 2019; Teixeira et al., 2024), and electricity price forecasting (Belenguer et al., 2025; Nowotarski and Weron, 2018; Weron, 2014). Forecasting for residential households is particularly crucial, for instance for designing home energy management systems (HEMS). HEMS monitor and control household energy flows to optimize energy consumption (load), reduce electricity costs, and improve user comfort (Gomes et al., 2022; Raza et al., 2024). However, forecasting at the household level presents unique challenges, e.g.,:

- Individual households exhibit highly variable and heterogeneous load profiles, shaped by factors such as user behavior, comfort factors, and physiological factors (Meyer et al., 2024; Nilsson et al., 2018).
- Increase in the use of distributed energy resources such as rooftop photovoltaic (PV) panels, electric vehicles (EVs), and residential storage further increases uncertainty in electricity profiles (Meyer et al., 2024; Zhou et al., 2022).

These challenges, combined with the limited availability of household data due to privacy considerations, highlight the need for accurate forecasting models which are robust even with limited data. Over the past decades, several forecasting models have been proposed for energy systems, partially thanks to the increase in computational power and data availability. Early works predominantly relied on classical statistical time series models (Brockwell and Davis, 1991; Chatfield, 2000; Hyndman and Athanasopoulos, 2018). For instance, for a long time, autoregressive (AR), moving-average (MA), and ARMA, a combination of AR and MA, which follows the Box-Jenkins framework (Box and Jenkins, 1971), were considered the standard methods (Makridakis and Hibon, 1997). These methods use autocorrelation within historical data and only require the target series as input. Exponential smoothing (Gardner, 1985) and Holt-Winters (Chatfield, 1978) seasonal methods were also widely adopted because of their simplicity and data efficiency. The strengths of statistical models lie in their interpretability and ability to produce accurate forecasts with limited data in comparison to deep learning (DL) models. However, their linear assumptions and reliance on stationarity restrict their capability to capture non-linear or complex temporal correlations, especially in rapidly changing energy dynamics (Akhtar et al., 2023; Henley et al., 2020; Makridakis et al., 2018). Owing to improved computational power, machine learning (ML) models such as decision trees (Quinlan, 1986), random forests (Breiman, 2001) and support vector regression (Smola and Schölkopf, 2004) became popular for load forecasting (Vivas et al., 2020). These approaches can learn non-linear relationships and integrate diverse input features. While classical ML models scale relatively well with increased dataset size, they can encounter difficulties with long temporal dependencies (Kontopoulou et al., 2023; Makridakis et al., 2023; Masini et al., 2023).

The DL revolution significantly influenced the direction of time series forecasting. Earlier DL frameworks employed multi-layer perceptrons (MLPs) (Bishop, 1995), which can model non-linear functions and automatically learn feature representations. Although simple MLPs lack temporal memory, they paved the way for more advanced architectures such as recurrent neural networks (RNNs) (Rumelhart et al., 1986), and their variants, long short-term memory (LSTM) (Hochreiter and Schmidhuber, 1997) and gated recurrent unit (GRU) (Cho et al., 2014). These architectures are capable of learning long-range temporal patterns and have demonstrated high accuracy in forecasting across various energy applications (Vivas et al., 2020; Meyer et al., 2024; Negash et al., 2025). Their advantages include flexible modeling of temporal dynamics and multivariate inputs. However, they require substantial training and data, and they have reduced interpretability. Convolutional neural networks (CNNs) (LeCun et al., 1989b), which gained attention in computer vision tasks (Bhatt et al., 2021; Mumuni and Mumuni, 2021; Zhao et al., 2024), have also been adapted for time series forecasting (Aouad et al., 2022; Jurado et al., 2023; Nguyen-Duc et al., 2024; Ullah et al., 2025). CNN-based models can extract important features from short to long time windows, and their ability to parallelize often results in faster training compared to RNNs. However, CNNs require careful design of the receptive field to adequately cover forecasting horizons, and very deep architectures are prone to overfitting when there is limited data (Song et al., 2024).

Transformer-based models, originally getting popular in the field of natural language processing (Patwardhan et al., 2023; Zhang and Shafiq, 2024), have shown their capability in accurate time series forecasting (Zhao et al., 2026). By employing self-attention (Vaswani

List of Abbreviations

ACF	Autocorrelation Function
AR	Autoregressive
ARIMA	Autoregressive Integrated Moving Average
ARMA	Autoregressive Moving Average
BESS	Battery Energy Storage System
BRD	Belgian Residential Data
BiTCN	Bidirectional Temporal Convolutional Network
CNN	Convolutional Neural Network
DL	Deep Learning
DeepNPTS	Deep Non-Parametric Time Series
DeepAR	Deep Autoregressive Recurrent Neural Network
DER	Distributed Energy Resources
DM	Diebold–Mariano
DSO	Distribution System Operator
EV	Electric Vehicle
GBDT	Gradient Boosting Decision Tree
GRU	Gated Recurrent Unit
HEMS	Home Energy Management System
iTransformer	Inverted Transformer
KNN	K-Nearest Neighbors
LCL	Low Carbon London
LSTM	Long Short-Term Memory
MA	Moving Average
MAE	Mean Absolute Error
ML	Machine Learning
MLP	Multi-Layer Perceptron
MQ-CNN	Multi-horizon Quantile Convolutional Neural Network
MQ-RNN	Multi-horizon Quantile Recurrent Neural Network
N-BEATS	Neural Basis Expansion Analysis for interpretable Time Series
NHITS	Neural Hierarchical Interpolation for Time Series
PACF	Partial Autocorrelation Function
PatchTST	Patch Time Series Transformer
PV	Photovoltaic
RNN	Recurrent Neural Network
RMSE	Root Mean Squared Error
STL	Seasonal-Trend-Loess
SFH	Single-Family Household
TCN	Temporal Convolutional Network
TFT	Temporal Fusion Transformer
TSFM	Time Series Foundation Model
TiDE	Time-series Dense Encoder
TimeGPT	Time Series Generative Pre-trained Transformer
Timer-XL	eXtra Long pre-trained Time-series Transformer
TimesFM	Time Series Foundation Model
TimeXer	Time Series Transformer with eXogenous variables
WPuQ	Wind Solar Heat Pump District
ZHC	Zonnedael Household Consumption

et al., 2017) mechanisms instead of recurrence, transformers capture dependencies across temporal steps more effectively than RNNs. The transformer architecture paved the way for foundation models, which are large-scale models pre-trained on massive datasets, and have since led to a significant paradigm shift across various research areas. Inspired by breakthroughs of BERT (Devlin et al., 2018) and GPT (Radford et al., 2018) in natural language processing, time series foundation models (TSFMs) were trained on massive, diverse time series datasets for tasks such as forecasting and anomaly detection (Garza et al., 2024). These models support zero-shot or few-shot learning, providing accurate forecasts on unseen data without re-training. Their primary advantage is data efficiency, as TSFMs utilize generic temporal patterns learned across millions of data points, making them particularly suited to contexts like household-level forecasting tasks, where historical data is limited (Meyer et al., 2024; Fu et al., 2026). However, these models are large and generally lack interpretability; their performance also depends on the relevance of pre-training data to the target forecasting tasks. Nonetheless, their ability to perform zero-shot forecasting represents a noteworthy advancement.

This paper differentiates itself from prior surveys by reviewing a wide spectrum of forecasting models, from classical statistical approaches to the state-of-the-art (SOTA) foundation models. We benchmark these methods across three household energy production and consumption tasks, namely load, PV, and battery energy storage system (BESS) as shown in Table 1. Previous surveys have either typically focused on a limited number of forecasting tasks or did not consider some of the model architectures included in this paper. For instance, while Qiu et al. (2024), Wang et al. (2024a) and Hasan et al. (2025) considered most of the statistical, ML, and deep learning models, they did not include foundation models. Additionally, most benchmarking efforts were limited to load forecasting, only Pandžić and Capuder (2023) and Qiu et al. (2024) presented benchmarks on PV energy data. To the best of our knowledge, no prior work surveyed or benchmarked all major forecasting architectures from statistical to ML, DL, and foundation models. In this paper, we present a comprehensive benchmark and survey of these models across three target forecasting tasks. By doing so, we reveal the evolving trade-offs among approaches in terms of data requirements, interpretability, scalability, and robustness, and underscore how foundation models introduce new capabilities such as transfer learning and zero-shot forecasting. The main contributions can be summarized as follows:

- **Comprehensive Benchmark and Survey:** We provide an extensive review that spans from classical statistical models to modern foundation models, integrating them into a unified benchmarking framework for consistent performance comparison.
- **Beyond Load Forecasting:** We extend the evaluation scope beyond load forecasting to include rooftop PV energy generation and BESS charging/discharging profiles, capturing the performance of the forecasting models across distributed energy resources.
- **A Paradigm Shift:** The rapid advancement of TSFMs introduces a potential paradigm shift in energy forecasting, driven by their ability to perform zero-shot predictions even in low-data scenarios. This survey is the first to integrate TSFMs into an extensive benchmarking framework in energy systems.

2. Preliminaries

In this section, we review some basic terminologies and concepts related to time series forecasting. We start by giving a definition of a time series and looking into different classes. Thereafter, we review time series decomposition, one of the most fundamental concepts that have influenced how forecasting models are designed. We also give a concise overview of stationarity, temporal dependence, and conclude by defining what time series forecasting entails.

2.1. Time series

A *time series* $X \in \mathbb{R}^{T \times N}$, is a set of observations collected over a specific period T (Brockwell and Davis, 1991). Here, N is the dimension of the time series, which represents the number of variables observed over a time period. When $N = 1$, the time series is termed *univariate*, and if $N > 1$, it is classified as *multivariate*. In the context of energy systems, such observations include energy consumption, PV energy generation, or battery operating patterns. Time series can be classified as discrete-time or continuous-time, depending on whether the observations are recorded at fixed time intervals or continuously over a time period, respectively (Brockwell and Davis, 1991, 2016).

Table 1
Comparison between our paper with recent related studies in terms of forecasting tasks and forecasting model coverage. The *Forecasting tasks* column specifies the tasks evaluated in the benchmarking process.

Reference	Forecasting tasks			Forecast model category					
	Load	PV	BESS	Statistical/ML	MLP	Recurrent	Convolutional	Transformer	Foundation
Pandžić and Capuder (2023)		✓		✓		✓	✓		
Wazirali et al. (2023)				✓	✓	✓	✓		
Meyer et al. (2024)	✓								✓
Qiu et al. (2024)	✓	✓		✓	✓	✓	✓	✓	
Wang et al. (2024a)	✓			✓	✓	✓	✓	✓	
Dong et al. (2025)					✓	✓	✓	✓	
Hasan et al. (2025)				✓	✓	✓	✓		
Musbah and Elsaraiti (2025)	✓			✓		✓	✓		
O'Connor et al. (2025)				✓	✓	✓	✓	✓	
Ugbehe et al. (2025)				✓	✓	✓	✓		
Our survey	✓	✓	✓	✓	✓	✓	✓	✓	✓

2.2. Classical decomposition

The initial step in time series analysis involves visualizing the data and its components. Time series data can be decomposed into distinct components that capture different patterns over time. Using classical decomposition, time series can be expressed as:

$$X = \mathcal{T} + \mathcal{S} + \mathcal{R}, \tag{1}$$

where $\mathcal{T}$ represents the underlying trend, $\mathcal{S}$ denotes the seasonal component, and $\mathcal{R}$ corresponds to the residual component. The pattern of these components varies with the nature of the data; hence, they should be properly understood when designing time series forecasting models (Brockwell and Davis, 2016; West, 1997).

The trend component reflects the general progression of the time series, which may result from factors such as economic growth, technological advancement, or population changes (Stock and Watson, 1988). Therefore, capturing the trend is vital to avoid biased forecasts that fail to reflect underlying changes. The seasonal component shows the regular and predictable patterns that recur over known periods, such as hours, days, or months. This is particularly an important component in energy systems because the patterns within the electricity data vary depending on fixed periods such as the time of the day, month, or season. The residual component captures the random noise, which is generally assumed to be stationary. If the variations of these components demonstrate non-stationarity, i.e., they appear to increase or decrease over time, a preliminary transformation such as differencing is required when using conventional statistical models such as Autoregressive Integrated Moving Average (ARIMA) (Brockwell and Davis, 2016).

Fig. 1 shows household consumption data and its components obtained using the Seasonal-Trend method based on Loess (STL) (Cleveland et al., 1990). This STL decomposition plot shows the original time series and its decomposition into trend, seasonal, and residual components. The original series shows high variability and repeated spikes, corresponding to daily peaks in energy usage. The trend component captures the long-term movement and shows a smooth, periodic increase and decrease over time. The seasonal component isolates regular short-term fluctuations, with consistent periodicity across the time series. Finally, the residual component represents the irregular or random variations that are not explained by the trend or seasonality.

2.3. Stationarity

In time series forecasting, stationarity is an essential prerequisite, as some classical models assume that the underlying process does not exhibit trends or seasonal variations over time. A stochastic process $\{X_t\}$ over a time period t is said to be stationary if its first and second moments are invariant with time (Brockwell and Davis, 2016). Specifically, this means that the mean:

$$\mu_X(t) = \mathbb{E}\left[X_t\right], \tag{2}$$

is independent of t, and the covariance:

$$\begin{aligned}\gamma_X(h) &= \mathrm{Cov}(X_{t+h}, X_t) \\ &= \mathbb{E}\left[(X_{t+h} - \mu_X(t+h))(X_t - \mu_X(t))\right],\end{aligned} \tag{3}$$

is independent of t for each time-shift h. This definition is commonly used; however, it is considered a weak case. For strict stationarity, a time series $\{X_t, t = 0, \pm 1, \ldots\}$ should satisfy the condition that the joint distribution of $(X_1, \ldots, X_n)$ is the same as that of $(X_{1+h}, \ldots, X_{n+h})$, for all integers h and $n > 0$. A particularly important special case of a stationary process is white noise. A white noise sequence $\{\varepsilon_t\}$ consists of uncorrelated random variables with zero mean $\mathbb{E}[\varepsilon_t] = 0$ and constant variance $\mathrm{Var}(\varepsilon_t) = \sigma^2 < \infty$.

2.4. Autocorrelation and temporal dependence

The autocorrelation function $\rho_X(h)$ is a fundamental statistical measure used to assess and quantify the linear dependence within a time series at various time lags. The autocovariance (covariance) measures the extent to which the values of the series at time points t and $t + h$ are related. The autocorrelation function (ACF):

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)} = \mathrm{Cor}(X_{t+h}, X_t), \tag{4}$$

normalizes this measure by dividing the autocovariance by the variance at lag zero (Brockwell and Davis, 2016). These measures provide insight into the temporal dependencies in the data; for instance, it can be assumed that $\gamma_X(h) \to 0$ as $|h| \to \infty$, reflecting the principle that long-term dependencies weaken over time.

In time series forecasting with finite order statistical models like AR and MA, the empirical ACF and partial autocorrelation function (PACF) are employed as measures in model identification. These functions provide insight into the underlying dependency structure and help determine the appropriate model class and order. For instance, if the PACF displays a sharp cutoff after lag p while the ACF decays gradually, this suggests that an autoregressive model of order p, denoted AR(p), may be appropriate. Conversely, if the ACF cuts off after lag q while the PACF tails off, it indicates a moving average model of order q, denoted MA(q) (Brockwell and Davis, 2016).

2.5. Time series forecasting

Time series forecasting is the process of estimating future values of a temporal sequence based on its historical behavior (Brockwell and Davis, 2016). This involves identifying and learning underlying aforementioned features in the data, such as trends, seasonality, and temporal dependence, to produce accurate forecasts. More generally, given a time series $\{X_t\}$, the goal is to predict future values X_{t+h} provided the available historical data $\{X_u : u \leq t\}$. Hence, the fundamental forecasting task can be represented as computing the

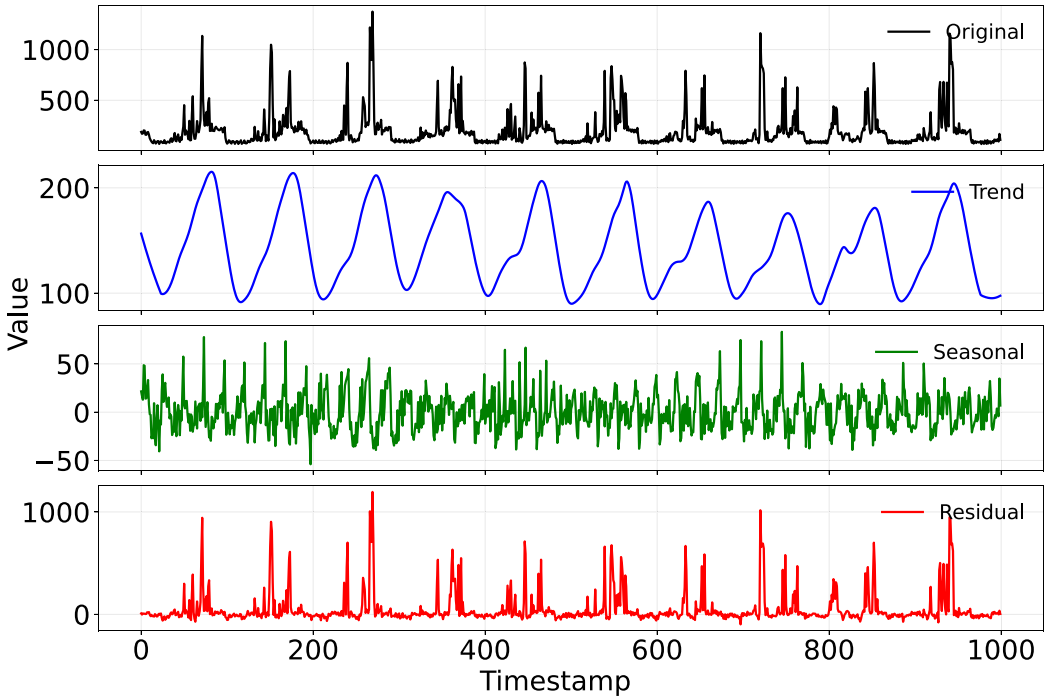


Fig. 1. STL decomposition of a residential electricity consumption time series into trend, seasonal, and residual components.

conditional expectation of the future observation, given the past, as follows:

$$\hat{X}_{t+1} = \mathbb{E}[X_{t+1} \mid X_t, X_{t-1}, \dots], \tag{5}$$

which can be seen as an optimal predictor that minimizes the forecast errors (Shumway and Stoffer, 2017).

Time series forecasting methods can be classified into univariate and multivariate approaches (Chatfield, 2000). Univariate methods rely solely on the present and historical values of the target time series, possibly incorporating a deterministic component like a linear trend. In contrast, multivariate methods generate forecasts based at least in part on additional time series variables referred to as predictor or explanatory variables and may involve systems of equations when the variables exhibit joint dependence. It is important to highlight the difference between a forecasting method and a forecasting model. A forecasting method refers to the procedure used to generate forecasts from available data, whereas a forecasting model is a mathematical framework that is capable of describing the behavior of the time series and can range from statistical approaches to those based on transformer architecture. The forecasting type can be direct, where all future steps are predicted in a single pass, or recursive, where predictions are generated one step at a time using the previous outputs, as illustrated in Fig. 2.

3. Methodology

This section outlines our methodological framework employed to conduct a comprehensive benchmarking survey of time series forecasting methods in energy systems. The primary objective is to evaluate the performance of several forecasting algorithms under a fair setup. The study mainly focuses on the systematic evaluation of the forecasting models using the default model parameters, with a limited hyperparameter search.

3.1. Design and scope

The survey evaluates 30 different time series forecasting models from the statistical, ML, DL, and zero-shot foundation model categories, as shown in Fig. 3. For statistical models, we selected conventional approaches that pioneered research in time series forecasting and have been widely used for a long time. For the other categories, we selected SOTA models proposed between 2016 and 2025. The selection of the models was intended to provide a representative and balanced benchmark rather than an exhaustive inclusion of all available methods. Moreover, we aim to assess the possibility of adopting previously proposed methods in practical applications, such as the forecasting stage of HEMS design. The model selection was guided by (i) empirical performance in prior benchmarking studies summarized in Table 1, including the most recently proposed TSFMs available at the time of writing; (ii) methodological diversity to capture different modeling paradigms; and (iii) the availability of stable and reproducible implementations.

The benchmarking process was conducted across three main household energy forecasting tasks, namely, power generation from rooftop PV systems, battery charging and discharging pattern prediction, and aggregated consumption. For each task, we adopted a consistent evaluation protocol, enabling direct comparisons across various models. The primary focus was on short-term forecasting scenarios. To ensure a fair comparison, all models were used with their standard default configurations as provided in their respective official papers, repositories, or widely used implementations. No hyperparameter tuning or architecture modifications were performed beyond necessary adjustments for input dimensions and the stopping criterion, such as the number of epochs, where necessary. This study highlights the out-of-the-box performance of the models and provides forecasting errors across multiple energy forecasting tasks with limited bias. It is important to note that fine-tuning and adjusting the models for domain-specific tasks may improve forecasting accuracy.

3.2. Epoch selection for deep learning models

We conducted preliminary experiments to determine the number of epochs for DL models. Specifically, we selected two SOTA models, Informer (Zhou et al., 2021) and PatchTST (Nie et al., 2023), and evaluated increasing training durations: 10, 20, 30, 50, 100, 200, and 500 epochs, using the load, PV, and battery profile data. The experiments showed that forecast error generally decreased up to around

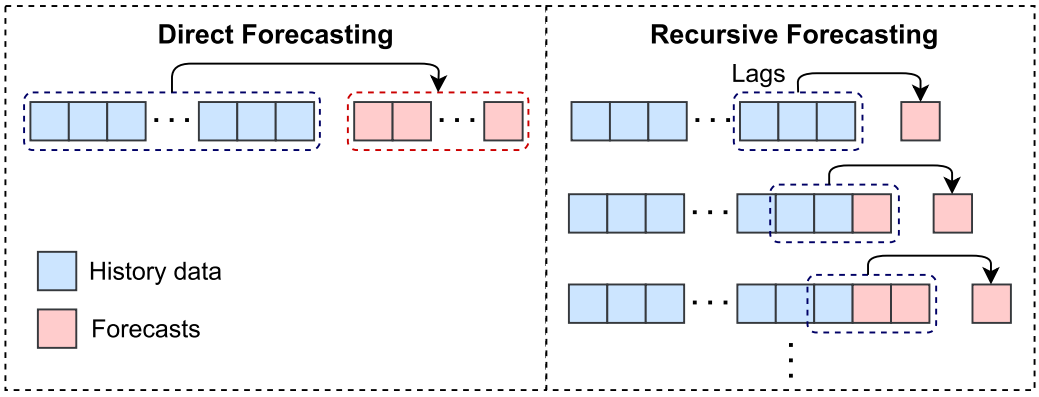


Fig. 2. Comparison of forecast types: direct forecasting predicts each future horizon independently, while recursive forecasting iteratively uses prior predictions as inputs for subsequent forecasts.

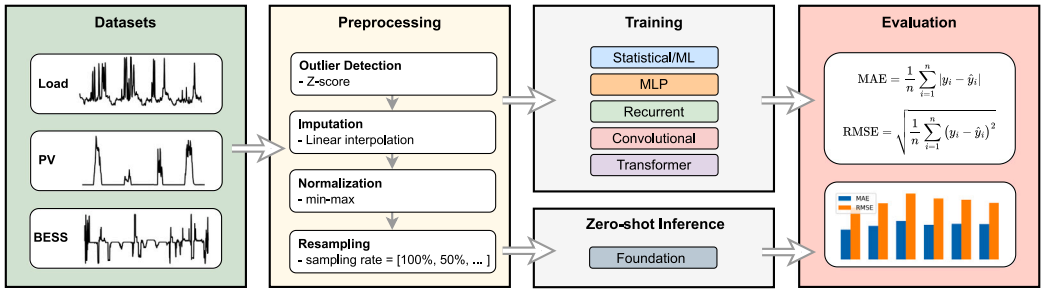


Fig. 3. Overview of our benchmarking methodology.

100 epochs, beyond which improvements were either insignificant or diminished. As a result, 100 epochs were chosen as a standard training duration for benchmarking all DL models. This decision balances performance and computational efficiency while ensuring consistency within the evaluation pipeline.

3.3. Zero-shot inference for foundation models

Foundation models have significantly influenced research in time series forecasting due to their ability to perform zero-shot inference. We use these models without any fine-tuning or task-specific adaptation. To evaluate the potential of large-scale pre-trained TSFMs, we included 5 models proposed between 2023 and 2025 in the survey using zero-shot inference. Forecasts were generated directly from the pre-trained model outputs using standardized inputs. We evaluated the zero-shot forecasting performance of TSFMs and compared them with that of trained statistical, ML, and DL models as shown in Fig. 3.

4. Forecasting models

For each of the models used in our benchmarking framework, we first briefly present the model and then discuss its advantages and disadvantages. Fig. 4 presents the timeline of these forecasting models while Table 2 summarizes the discussion of the models.

4.1. Statistical and machine learning models

Statistical and conventional ML models encompass both naive heuristic approaches and established parametric methods. Naive Drift (Chatfield, 2000; Hyndman and Athanasopoulos, 2018) allows the forecasts to either increase or decrease over time, with the rate of change, referred to as the "drift", set to the average change observed in the historical data plus some random noise. Likewise, Naive MA (Brockwell and Davis, 2016; Chatfield, 2000) forecasts future values by calculating the average of the most recent k observations. An AR model, on the other hand, forecasts future values based on a linear combination of its past values. The AR model captures the temporal dependencies in the data and is useful for time series with significant autocorrelation. ARIMA (Brockwell and Davis, 2016; Chatfield, 2000) is a parametric method that combines both AR and MA components, along with an integrated (I) term that accounts for non-stationarity by differencing the series.

K-Nearest Neighbors (KNN) regression is an instance-based non-parametric approach that forecasts by finding the k closest historical sequences to the current context and averaging their next values (Mitchell, 1997). KNN has been used in several time series tasks such as forecasting stock prices (Zhang et al., 2017), electricity demand (Al-Qahtani and Crone, 2013; Hnin et al., 2024), and electricity prices (Ali et al., 2019). Gradient Boosting Decision Tree (GBDT) is a popular ML algorithm known due to its high efficiency, predictive accuracy, and interpretability (Friedman, 2001). However, many GBDT models struggle with efficiency and scalability in high-dimensional settings, as they must scan all data instances to compute the information gain for every possible split point, which is time-consuming. LightGBM (Ke et al., 2017), on the other hand, improves efficiency by excluding a large portion of instances with small gradients and using only the remaining ones to estimate information gain.

4.2. Multi-layer perceptron-based models

MLP-based models for time series forecasting employ feedforward neural networks to map fixed windows of past data directly to future predictions. These architectures often use residual connections or specialized output bases to capture complex temporal patterns (Ekambaram et al., 2023). Neural Basis Expansion Analysis for interpretable Time Series (N-BEATS) (Oreshkin et al., 2019) is a deep neural architecture that stacks fully connected layers into "blocks", each producing both a backcast (reconstruction of the input window) and a forecast.

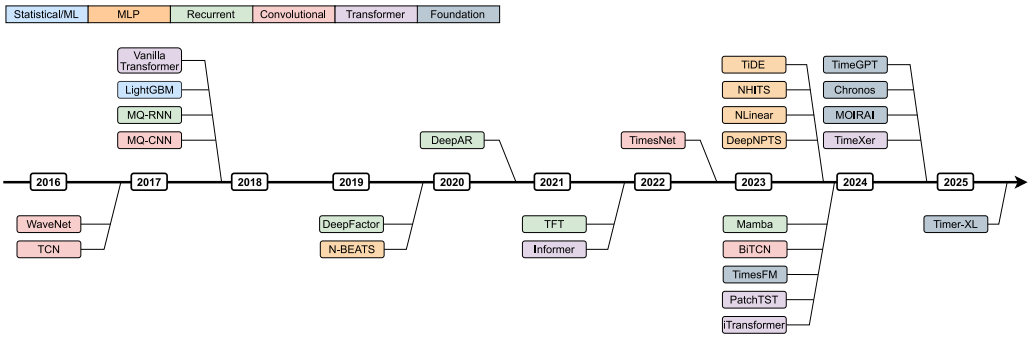


Fig. 4. Timeline illustrating the development of various forecasting models from 2016 to 2025.

Olivares et al. (2023) enhance the N-BEATS model by incorporating exogenous variables, resulting in N-BEATSx, enabling it to use additional external information from multiple sources of useful information to improve forecasting accuracy. Challu et al. (2023) also extended N-BEATS to develop Neural Hierarchical Interpolation for Time Series (NHITS) and introduced hierarchical interpolation and multi-rate pooling, allowing different blocks to specialize in distinct frequency components; for instance, some focusing on long-term trends and others on short-term variations. Nedić et al. (2025) combined N-BEATS with CNN for load forecasting.

Unlike N-BEATS and NHITS, which are feedforward networks without a dedicated encoder, Time-series Dense Encoder (TiDE) (Das et al., 2023a) is an encoder–decoder model. It first encodes the past observations and any known covariates into a latent representation using a dense network, and then a separate dense decoder network generates the forecast sequence. NLinear (Zeng et al., 2023) is a simple feedforward model that subtracts the last observed value from the input series, applies a linear transformation, and then adds the last value back to produce predictions. Essentially, NLinear learns a fixed matrix to map recent differences to future differences, making it effectively a linear regression model on the differenced series. Rangapuram et al. (2023) proposed Deep Non-Parametric Time Series (DeepNPTS), a non-parametric approach that learns the sampling distribution directly from the observed data and then continues sampling only from the observed values. The forecasts for each time series are then made by sampling from this learned distribution at the forecast time step.

4.3. Recurrent network-based models

Recurrent network-based models process time series sequentially using recurrent layers such as LSTM and GRU, often enhanced with dilations, attention mechanisms, or hybrid components to capture long-term temporal dependencies (Karpathy et al., 2015). Dilated RNN (Chang et al., 2017), i.e., an RNN variant with multi-resolution dilated recurrent skip connections, stacks RNN layers but adds dilations in time to capture multi-resolution patterns. This means that higher layers skip more time steps, allowing the network to see far in the past without extremely deep recurrence. Classical time series models often fall short in capturing complex patterns, while multivariate approaches struggle to scale effectively to large datasets. To overcome these challenges, Wang et al. (2019) proposed DeepFactor, a hybrid framework that combines deep neural networks with classical probabilistic models. Their core idea was to represent each time series as a combination of a shared global component and a series-specific local model. DeepAR (Salinas et al., 2020) is another probabilistic forecasting model that trains an autoregressive RNN on a large set of related time series. At each step, it takes as input the current covariates, the previous target value, and the previous hidden state to update its hidden representation and produce the parameters of a predictive distribution. During forecasting, the model uses actual historical values up to the forecast start time, then generates future values by sampling from the predicted distribution and feeding each sample back for the next step. Similar to DeepAR, Multi-horizon Quantile RNN (MQ-RNN) (Wen et al., 2017) is autoregressive, but it predicts multiple quantiles of the future distribution in one forward pass. It uses a "forking-sequences" training scheme that stabilizes multi-step learning. Specifically, the model encodes the history with an RNN and then decodes a block of future quantiles, outputting multiple quantile values per time step.

Temporal Fusion Transformer (TFT) (Lim et al., 2021) is a hybrid sequence-to-sequence model that integrates recurrent networks with attention mechanisms. Although it carries the name "Transformer", TFT is fundamentally a recurrent network-based approach, as it relies on LSTM layers to capture short-term temporal dynamics. It incorporates interpretable multi-head self-attention to model long-term dependencies, along with gating and variable selection modules to prioritize relevant features. In TFT, past inputs and covariates are processed through LSTM-based encoders, while future predictions are generated by a decoder that attends to the encoded representations. Mamba (Gu, 2023) is another architecture that draws inspiration from an established area, specifically control theory, by using linear state-space layers to propagate and mix hidden states over time. This design achieves near-linear time complexity with respect to sequence length, in contrast to the quadratic cost associated with self-attention mechanisms. Conceptually, Mamba functions as an efficient recurrent model, replacing traditional RNN cells with linear state-space modules to improve efficiency. S-Mamba (Wang et al., 2025) enhances this architecture by using a bidirectional Mamba layer to extract inter-variate correlations and a feedforward network to capture temporal dependencies more effectively. Zhao et al. (2025) proposed a model that combines symplectic geometry mode decomposition and the Mamba to enhance medium-term and long-term PV power forecasting accuracy.

4.4. Convolutional network-based models

Convolutional models for time series use causal and dilated convolutions to capture temporal dependencies efficiently. The final forecasts are typically produced by a simple output layer or decoder (LeCun and Bengio, 1998; LeCun et al., 1989a). A Temporal Convolutional Network (TCN) is a causal architecture using dilated 1D convolutions. It applies stacked convolutional layers with increasing dilation rates so that the receptive field grows exponentially with depth. The output sequence of each layer matches the input length due to causal padding, and finally, an MLP decodes the convolutional features into forecasts. Bidirectional TCN (BiTCN) (Sprangers et al., 2023) uses two temporal blocks, the "forward" dilated convolutions that encode any future-known covariates, and the "backward" one that encodes past values and covariates. These representations are then combined before a final dense layer produces the forecast.

Earlier approaches for modeling temporal variation relied on directly using raw 1D time series data (Franceschi et al., 2019; He and Zhao, 2019). WaveNet (Oord et al., 2016), originally developed for 1D audio data, has since been adapted for time series forecasting. It

Table 2
Summary of forecasting models including their methods, types, forecast type, main advantages, and limitations.

Model	Forecast method	Model type	Forecast type	Benefits	Limitations
KNN Regression (Mitchell, 1997)	Both	Instance-based	Both	Makes no strong assumptions about the data distribution and adapts to different temporal patterns by design.	Requires careful feature engineering, can be slow, and computationally intensive for high-dimensional inputs.
LightGBM (Ke et al., 2017)	Multivariate	Tree-based	Recursive	Computationally efficient, interpretable, handles large datasets, and captures non-linear relationships without heavy tuning.	Lacks inherent temporal understanding and relies on lag features to capture time dependencies.
ARIMA (Chatfield, 2000)	Univariate	Parametric	Recursive	Interpretable, suitable for stationary data, and can model trends and seasonality.	As a linear model, it struggles to model sudden changes or complex non-linear patterns.
Naive Drift (Chatfield, 2000; Hyndman and Athanasopoulos, 2018)	Univariate	Non-parametric	Direct	Quick to compute, serving as a useful benchmark for model performance.	Can be inaccurate if trends change or seasonality exists because it ignores all historical data except the first and last points.
Naive MA (Brockwell and Davis, 2016; Chatfield, 2000)	Univariate	Non-parametric	Recursive	Performs reasonably well for stationary time series without clear trends or seasonality.	Cannot capture trends or seasonality beyond its fixed window size.
NHITS (Challu et al., 2023)	Univariate	Stacked MLP blocks	Direct	Computationally efficient, and the multi-rate design effectively captures both short- and long-term patterns.	Requires a fixed-length input window, which can limit flexibility when dealing with irregularly sampled time series.
N-BEATS (Oreshkin et al., 2019)	Univariate	Stacked MLP blocks	Direct	It is interpretable, readily applicable to a broad range of target domains without requiring modifications, and efficient to train.	Fixed input window size and limited explicit temporal modeling.
TiDE (Das et al., 2023a)	Univariate	Encoder-decoder	Direct	Simple and fast model. It effectively handles covariates and non-linearities.	May struggle with long-range dependencies and if series dynamics are very complex.
NLinear (Zeng et al., 2023)	Univariate	Linear	Direct	Simple, fast, and effective under distribution shifts.	Being linear, it cannot capture non-linear patterns, seasonality, or complex trends.
DeepNPTS (Rangapuram et al., 2023)	Univariate	Probabilistic	Direct	Simple and data-driven, using information across multiple series. Robust and effectively handles seasonality and trends.	Computational complex and relies on large amounts of historical data to accurately learn the sampling distribution.
DeepFactor (Wang et al., 2019)	Multivariate	Probabilistic	Recursive	Combines deep networks (global) for complex patterns with classical (local) models for handling uncertainty.	Implementation is complex, and modeling local and global models increases computational cost.
DeepAR (Salinas et al., 2020)	Univariate	Probabilistic	Direct	Effectively handles uncertainty and widely-varying scales.	Slow for long forecast horizons and performance may reduce on large datasets.
TFT (Lim et al., 2021)	Univariate	Sequence-to-sequence	Direct	Interpretable multi-horizon forecasting and captures both short- and long-term temporal patterns for diverse input types.	Computationally complex and its interpretability, though improved, remains less intuitive than of simpler models.
MQ-RNN (Wen et al., 2017)	Both	Sequence-to-sequence	Direct	Robust performance in large-scale applications, handles both temporal and static covariates.	May struggle with long-range dependencies due to the limitations of recurrent architectures.
Mamba (Gu, 2023)	Both	State-space	Recursive	Handles long sequences efficiently with faster inference compared to attention-based models.	Can be complex to implement, and sparse or irregular time series can challenge its assumptions and degrade performance.
TCN (Lea et al., 2016)	Both	Encoder-decoder	Direct	Captures long-range dependencies, avoids RNN vanishing gradients, and convolutions can be computed in parallel.	May underperform in domain transfer if their receptive field is not large enough for the new context.
BiTCN (Sprangers et al., 2023)	Univariate	Encoder-decoder	Direct	Uses two TCNs to effectively encode future covariates, and past observations.	Still has fixed receptive fields and may struggle to capture long dependencies. Less flexible than fully recurrent models.
TimesNet (Wu et al., 2022)	Both	Encoder-only	Direct	Models complex seasonality and multiple frequencies efficiently by transforming the series into 2D tensors.	May struggle with irregularly sampled data and processing 2D structures requires more computational power.
WaveNet (Oord et al., 2016)	Both	Probabilistic	Recursive	Models long-range dependencies with relatively few layers and can produce highly accurate forecasts.	Training can be slow and memory-intensive due to autoregression.
MQ-CNN (Wen et al., 2017)	Both	Sequence-to-sequence	Direct	CNNs can be parallel and capture local patterns, providing a fast alternative to RNNs for quantile forecasting.	Has a fixed receptive field, hence it may struggle to capture long-range temporal dependencies.

(*continued on next page*)

Table 2 (*continued*).

Informer (Zhou et al., 2021)	Both	Encoder-decoder	Direct	Handles long sequences efficiently. Fast inference due to its generative decoder which gives forecasts in one operation.	May lose some contextual information due to the ProbSparse attention which full attention might capture.
PatchTST (Nie et al., 2023)	Both	Encoder-only	Direct	High accuracy with much less computation by patching the input. Supports self-supervised pre-training.	Ignores cross-channel correlations by default due to its channel-independent design.
iTransformer (Liu et al., 2023)	Multivariate	Encoder-only	Direct	Enhances performance, generalization, and effective use of long lookback windows.	Neglecting temporal-wise attention and relying only on linear projection may not fully capture temporal dependencies.
Vanilla Transformer (Vaswani et al., 2017)	Univariate	Encoder-decoder	Direct	Can effectively capture long-range dependencies due to the attention mechanism.	Has $O(L^2)$ complexity, making long series computationally expensive.
TimeXer (Wang et al., 2024b)	Both	Encoder-only	Direct	Explicitly models the interaction of external (exogenous) features with the target series, improving accuracy.	Attention structure used can disrupt temporal order, possibly reducing accuracy in modeling sequential dependencies.
TimeGPT (Garza et al., 2024)	Both	Encoder-decoder	Direct	Easy to perform zero-shot forecasting and fine-tune, and also supports anomaly detection.	Proprietary model with limited details of the architecture. Requires enormous data to train.
TimesFM (Das et al., 2023b)	Both	Decoder-only	Direct	Smaller (200M parameters) than other TSFMs but still achieves high zero-shot performance.	Decoder-only design limits the ability to explicitly encode complex historical patterns, which may reduce accuracy.
MOIRAI (Woo et al., 2024)	Both	Encoder-only	Direct	A "universal forecaster" with high zero-shot performance, and supports long horizons and diverse distributions.	Relies on a heuristic multi-patch design for cross-frequency learning and has limited scalability to high-dimensional data.
Chronos (Ansari et al., 2024)	Both	Encoder-decoder	Direct	Strong zero-shot forecasting by pretraining on diverse tokenized time series, simplified pipelines while achieving high accuracy.	Relies on synthetic data that may not fully capture real complexities due to limited large and diverse real-world datasets.
Timer-XL (Liu et al., 2025)	Both	Decoder-only	Recursive	Effectively models both intra- and inter-series dependencies across varying-length contexts.	The autoregressive design can cause error accumulation. High computational complexity for high-dimensional data.

uses stacks of dilated causal convolutions, where each layer exponentially expands the receptive field. Gating mechanisms control which features propagate through the network. WaveNet essentially acts as a convolutional encoder of the input sequence, followed by a linear output layer to predict the next value. Another model that uses 1D convolution layers is Multi-horizon Quantile CNN (MQ-CNN) (Wen et al., 2017), which is a convolutional counterpart to MQ-RNN. It applies a convolutional sequence-to-sequence architecture to output multiple quantiles for each future step. A disadvantage of directly using 1D data for temporal variation modeling is the difficulty in learning the complex temporal patterns. To address this problem, TimesNet (Wu et al., 2022) was proposed, which transforms a 1D series into a 2D structure to capture multiple periodicities. It first embeds the input series and applies a Fast Fourier Transform to decompose it into multiple frequency bands. Each band is then reshaped into a 2D tensor, and a 2D convolutional block processes these tensors, capturing both intra-period and inter-period variations.

4.5. Transformer models

Transformer-based models utilize self-attention mechanisms on time series data, enabling them to capture long-range dependencies and inter-variable relationships more effectively than fixed-window or sequential models (Wen et al., 2022). Vaswani et al. (2017) first introduced the Transformer, an encoder–decoder architecture, commonly referred to as Vanilla Transformer in the literature to distinguish it from the several variants and adaptations that have since been proposed. The Transformer architecture consists of stacks of self-attention and fully connected layers in both the encoder and decoder. The core mechanism of this model is the scaled dot-product attention, which computes attention weights using queries Q, keys K, and values V as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{6}$$

where d_k is the dimension of the keys. Scaling by $\sqrt{d_k}$ prevents large dot-product values that can cause vanishing gradients in the softmax, thus improving training stability and efficiency.

Informer (Zhou et al., 2021), one of the transformer variants, retains the standard encoder–decoder architecture but introduces specialized self-attention to handle long sequences. It uses *ProbSparse* self-attention to reduce complexity from $O(L^2)$ to $O(L \log L)$ for L-length inputs/outputs, and a self-attention distillation technique to compress information. Liu et al. (2023) proposed inverted Transformer (iTransformer), an encoder-only framework that embeds the time points into variate tokens for attention-based multivariate correlation modeling, while a feed-forward network learns the nonlinear representations. Patch Time Series Transformer (PatchTST) (Nie et al., 2023), which is also an encoder-only model, was designed to enhance long-term multivariate time series forecasting. It divides each time series into subseries-level patches and enforces channel independence, i.e., each input token contains information from only a single time series (channel), without mixing features across variables. This patching strategy significantly reduces the effective input length to the attention mechanism while preserving local temporal structure. **Time** Series Transform**er** with e**X**ogenous variables (TimeXer) (Wang et al., 2024b) simultaneously uses patch-wise self-attention and variate-wise cross-attention, effectively reconciling information between the target series (endogenous) and external covariates (exogenous). It also learns global tokens to bridge exogenous and endogenous series, enabling the model to integrate external features effectively.

4.6. Foundation models

TSFMs are large pre-trained architectures, enabling zero-shot or transfer learning across diverse tasks, including forecasting (Liang et al., 2024). TimeGPT (Garza et al., 2024) is a large-scale model based on the Vanilla Transformer architecture (Vaswani et al., 2017),

pre-trained on over 100 billion time series data points, and supports zero-shot forecasting and anomaly detection. It generates forecasts by using a window of past observations and local positional encodings. It follows a multi-layer encoder–decoder architecture, incorporating residual connections and layer normalization. It also consists of a linear layer that transforms the decoder output into the forecasting window dimension. TimesFM (Das et al., 2023b), inspired by the patching strategy of PatchTST (Nie et al., 2023), employs a decoder-only architecture instead of the traditional encoder–decoder. It is trained autoregressively to predict the next patch based on all preceding patches within a context window, enabling flexible forecasting across variable-length inputs. Each input patch is transformed into a vector using a residual block, which is projected to the model dimension. This vector is then combined with positional encodings and fed into stacked transformer layers. The resulting output tokens are processed by a residual block to produce forecasts of length equal to the output patch.

Woo et al. (2024) introduced MOIRAI, a masked encoder-based universal time series forecasting transformer. To support the "any-variate" setting, they flatten multivariate time series into a single sequence and use a multi-patch-size input projection to embed the patches. A learnable mask token replaces patches within the forecast horizon. The resulting tokens are then decoded through a multi-patch-size output projection layer. Chronos (Ansari et al., 2024) represented time series as sequences of quantized tokens, which are processed using a transformer-based language model. The model was trained with cross-entropy loss on large public datasets. At inference time, Chronos autoregressively generates future tokens, which are then decoded back into numeric time series data. Liu et al. (2025) recently introduced Timer-XL, an extension of the Timer model (Liu et al., 2024) pretrained on over 1 billion time points, with a focus on long-context time series forecasting. They proposed a universal *TimeAttention* mechanism to capture fine-grained intra- and inter-series dependencies from flattened patches.

5. Experiments

In this section, we describe our experimental setup, including pre-processing steps, datasets, evaluation metrics, and details of experiments execution. The objective of our extensive experiments is to evaluate the performance of different forecasting models on short-term residential energy forecasting tasks for load, PV, and BESS charging/discharging profile data.

5.1. Experimental setup

For models that give deterministic forecasts, i.e., classical statistical/ML and zero-shot foundation models, we conducted a single run and directly report the results. The model hyperparameters are set to their default values as provided in the corresponding open-source repositories or official publications. For DL models, we executed each experiment 10 times, each with a different starting seed, and report the average forecast errors to ensure the precision of the results. For each time series $\{X_t\}_{t=1}^{T}$, we designed the training set as $\{X_t\}_{t=1}^{T-H}$ and the test set as $\{X_t\}_{t=T-H+1}^{T}$, where H is the forecast horizon. We evaluate multiple forecast horizons by varying the temporal resolution of the dataset through different sampling rates: $[100\%, 50\%, 33.33\%, 25\%]$. These sampling rates correspond to training and testing on the full dataset, selecting one data point out of every two, one out of every three, and one out of every four, respectively. For example, in a dataset with measurements recorded every 15 min, these sampling rates yield forecast horizons of $[192, 96, 64, 48]$ time steps, respectively.

5.2. Data pre-processing

The datasets underwent a standardized pre-processing pipeline to ensure comparability across models and domains. We employed a Z-score-based *outlier detection* method to remove statistical outliers from the data. For a time series x_t, the Z-score is computed as:

$$Z_t = \frac{x_t - \mu}{\sigma}, \tag{7}$$

where μ and σ are the mean and standard deviation of the time series, respectively. Data points with $|Z_t| > 3$ are flagged as anomalies and replaced via *imputation*. The missing observations were imputed using linear interpolation. For missing value x_t, the imputed value $\hat{x}_t$ is given by:

$$\hat{x}_t = x_{t_1} + \frac{x_{t_2} - x_{t_1}}{t_2 - t_1}(t - t_1), \tag{8}$$

where t_1 and t_2 are the indices of the nearest previous and next observed values. Lastly, we *normalized* all data to the range $[0, 1]$ using min–max normalization calculated as follows:

$$x'_t = \frac{x_t - x_{\min}}{x_{\max} - x_{\min}}, \tag{9}$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the series, respectively. This normalization ensures that all variables are on a comparable scale, enabling a fair comparison of forecasting errors of the models.

5.3. Evaluation metrics

We evaluated the forecasting performance of the models using the mean absolute error (MAE) and the root mean squared error (RMSE), defined as:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|, \tag{10}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}, \tag{11}$$

where y_i and $\hat{y}_i$ are the normalized ground truth and forecast values, respectively, and n is the number of data points in the test set. These metrics are widely used to evaluate forecasting accuracy in a fair way because they capture different aspects of model performance. The MAE measures the average magnitude of forecasting errors, offering interpretability and robustness to outliers. In contrast, the RMSE calculates the square root of the average squared differences between predicted and actual values, thereby penalizing larger errors more heavily. Consequently, the MAE is preferred when all errors are equally important, whereas the RMSE is more suitable when large deviations are particularly undesirable. Reporting both metrics provides a balanced assessment of model performance (Hyndman and Koehler, 2006; Willmott and Matsuura, 2005).

5.4. Datasets

We evaluated the models using the following electricity datasets from Belgium, Germany, the Netherlands, and the UK, all collected under real-world conditions. Experiments were conducted on a few selected customers, as shown in Table 3.

Wind Solar Heat Pump District (WPuQ) (Schlemminger et al., 2022): The dataset contains measurements of electric load profiles that provide insights into consumption patterns. It was collected within the framework of the research project between 2018 and 2020 in 38 single-family households (SFH) located in a district in Lower Saxony, Germany. The temporal resolution of the measurements ranges from 10 s to 60 min. The SFH 19 consumption dataset is used as the emulated BRD consumption data and follows the same training and forecasting horizons as the BRD dataset.

Table 3
Overview of electricity datasets utilized in this study where *Customer* represents the customer identifier in the dataset and *Length* refers to the number of data points.

Dataset	Customer	Forecasting task	Frequency	Length
WPuQ (Schlemminger et al., 2022)	SFH 19	Load	15 min	8736
BRD (Ruddick et al., 2024)	PV 1–4, BESS 1–4	Solar, Battery	15 min	8736
LCL (UK Power Networks, 2014)	LCL 33	Load	30 min	17,507
ZHC (Liander, 2014)	ZHC 8, 9, 43	Load	15 min	17,376

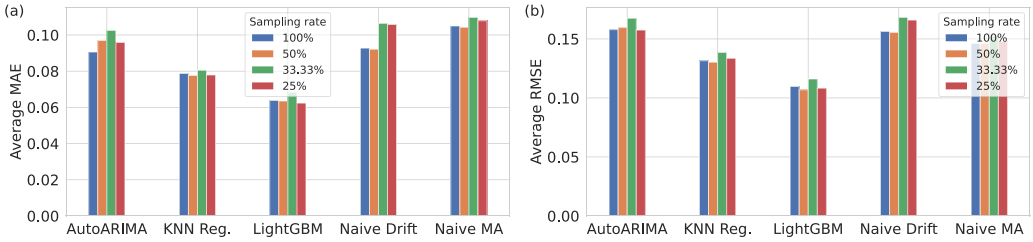


Fig. 5. Average forecasting errors of statistical and conventional ML models across all datasets.

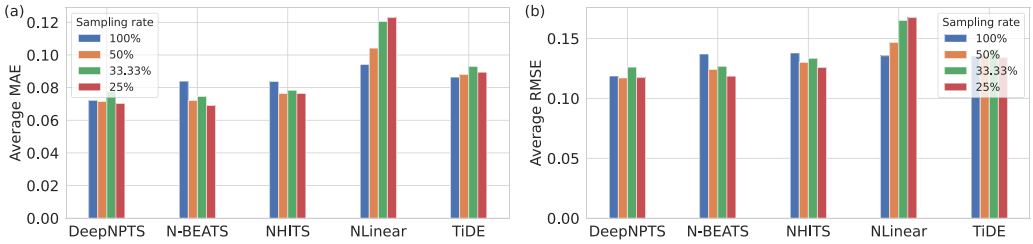


Fig. 6. Average forecasting errors of MLP-based models across all datasets.

Belgian Residential Data (BRD) (Ruddick et al., 2024): The data was collected from an experimental setup by Ruddick et al. (2024) in 2024 consisting of four physical replicas of a house electrical circuit, each outfitted with a BESS, a PV installation, EV charging sessions, and electric consumption emulated from the WPuQ dataset. All houses were connected to the grid and monitored by official digital meters from the local Belgian distribution system operator, recording both energy imported from and exported to the grid. In this work, we train the models using PV and BESS data collected from 1 January 2024 to 29 March 2024, and then generate forecasts for 30 and 31 March 2024.

Low Carbon London (LCL) (UK Power Networks, 2014): The dataset contains half-hourly energy consumption measurements for 5567 London households collected during the UK Power Networks LCL project from November 2011 to February 2014. A subgroup of about 1100 households participated in a dynamic time-of-use pricing trial in 2013, receiving high, low, or normal price periods communicated a day in advance via in-home display or text message, with the schedule included in the dataset. The remaining 4500 households were on a fixed rate. The pricing signals were designed to assess strategies for managing high renewable generation periods and mitigating stress on local distribution networks. In this work, we train the models using consumption data collected from 1 January 2013 to 27 December 2013, and then generate forecasts for the period 28 to 31 December 2013.

Zonnedael Household Consumption (ZHC) (Liander, 2014): The dataset contains actual consumption data from approximately 80 addresses in the Netherlands, with electrical measurements recorded at 15-minute intervals. The data was collected from smart meters with user consent for research purposes from May 1, 2012, to March 7, 2014. For this research, three customers were selected, identified as numbers 8, 9, and 43, corresponding to high, medium, and low energy consumption profiles, respectively. In this work, we train the models using consumption data collected from 1 January 2013 to 28 June 2013, and then generate forecasts for 29 and 30 Jun 2013.

6. Results

This section presents the forecasting performance of statistical, conventional ML, DL, and zero-shot TSFMs across multiple temporal sampling rates (100%, 50%, 33.33%, and 25%) corresponding to forecast horizons of [192, 96, 64, 48] time steps, respectively. For the WPuQ, ZHC, and BRD datasets, this corresponds to a 48-hour forecast horizon, while for the LCL dataset, it corresponds to a 96-hour forecast horizon. Reducing the sampling rate does not change the forecast horizon, it only decreases the number of forecasted data points within that horizon. The performance is evaluated for diverse targets, including household load (SFH 19, LCL 33, ZHC 8, ZHC 9, ZHC 43), solar (PV1–PV 4), and battery charging/discharging profiles (BESS 1–BESS 4). For each model and sampling rate, the average MAE and RMSE over these datasets are presented in Figs. 5–10, enabling a direct comparison of the forecasting accuracy of the methods in this study. We also present the normalized actual and forecast values at 100% sampling rate for load (ZHC 9), PV 1, and BESS 3 in Figs. 11–13, respectively, and the histograms of the best performing models from each class in Fig. 14. The numerical results are presented in Tables A.4–A.9 and additional visualizations of the normalized actual versus forecast values at 100% sampling rate are shown in Figs. A.18–A.27 in Appendix.

6.1. Discussion

Among statistical and conventional ML models, LightGBM consistently achieved the best accuracy across all domains and sampling rates as shown in Fig. 5. This shows that the gradient boosting trees effectively captured non-linear dependencies, yielding the lowest MAE and RMSE in most cases. KNN Regression offered the lowest RMSE value on SFH 19 at full sampling, but the RMSE performance became slightly poorer with reduced sampling. Naive Drift generally produced the highest errors because of its excessive fluctuations in the forecasts.

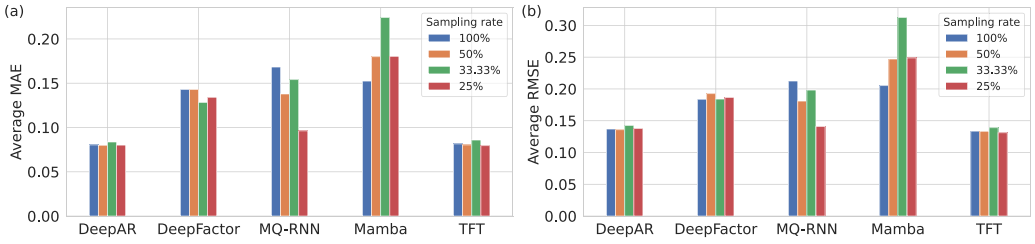


Fig. 7. Average forecasting errors of recurrent network-based models across all datasets.

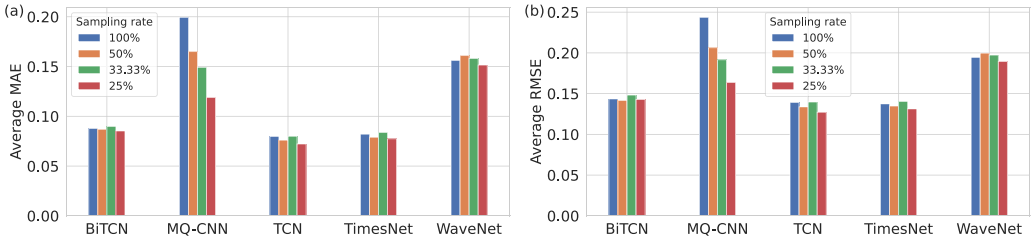


Fig. 8. Average forecasting errors of convolutional network-based models across all datasets.

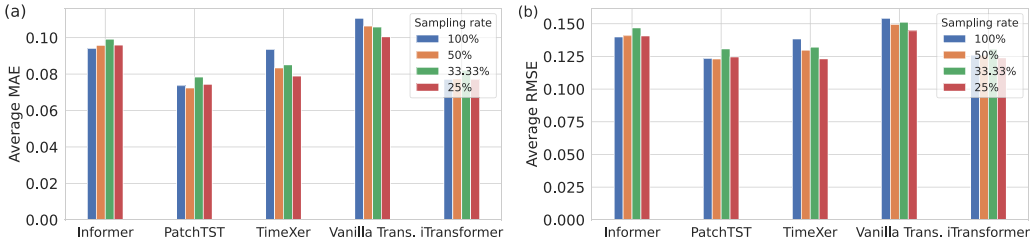


Fig. 9. Average forecasting errors of transformer-based models across all datasets.

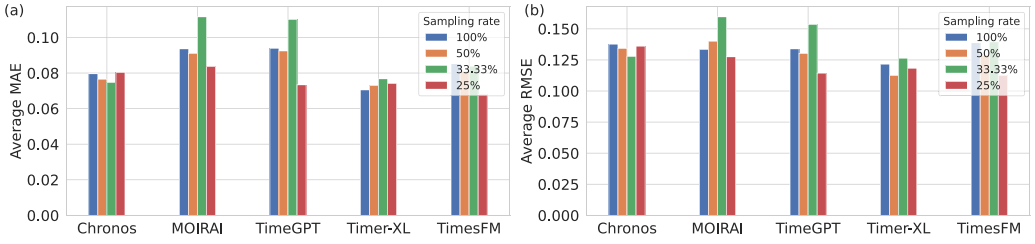


Fig. 10. Average zero-shot forecasting errors of foundation models across all datasets.

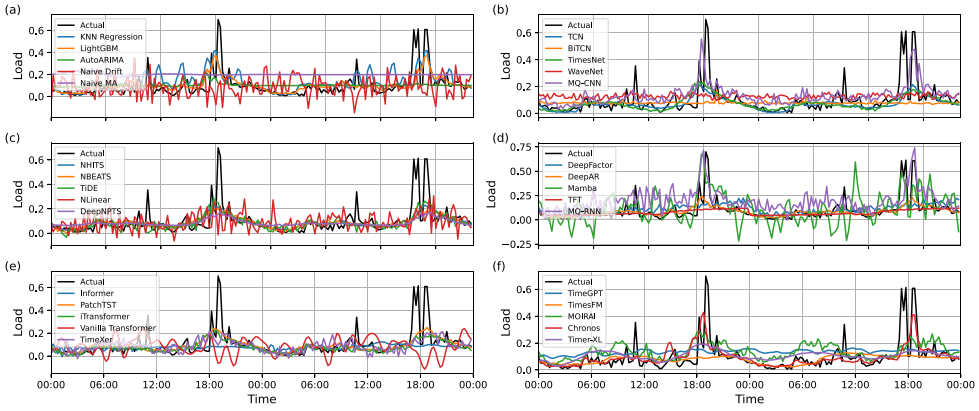


Fig. 11. Load forecasting results for customer 9 in the ZHC dataset.

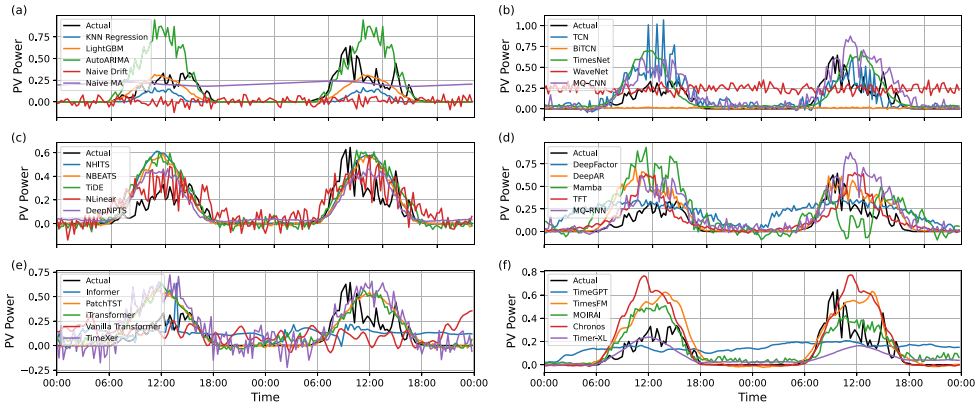


Fig. 12. PV generation forecast results for house 1 in the BRD dataset.

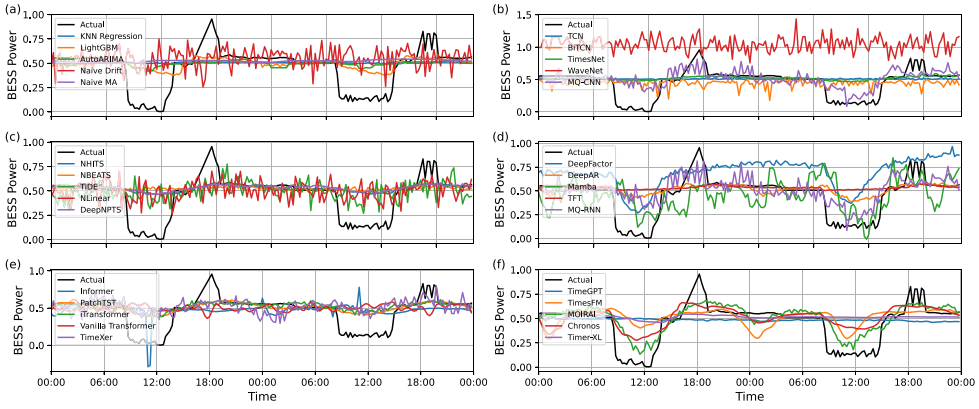


Fig. 13. BESS operation forecast results for house 3 in the BRD dataset.

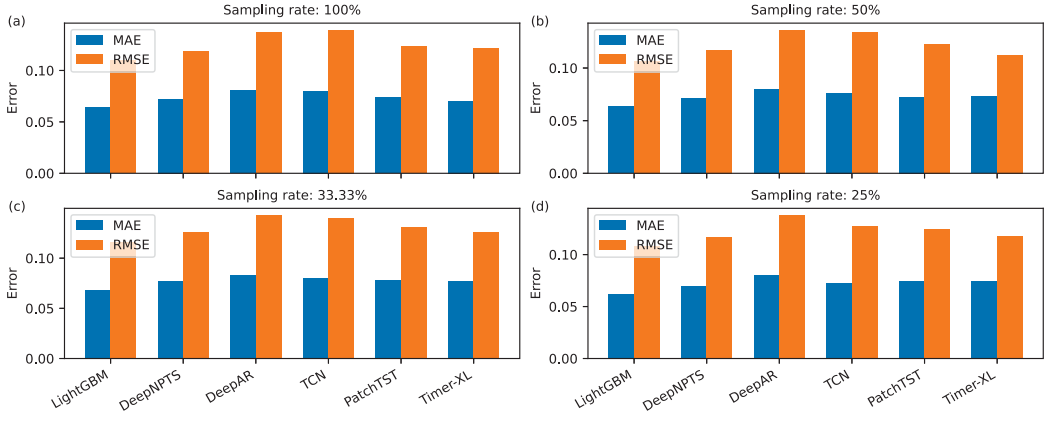


Fig. 14. Average forecasting accuracy of the best-performing models across all datasets.

Across all models, error increased as the sampling rate decreased; LightGBM showed the greatest stability, with minimal RMSE changes. For MLP-based models, DeepNPTS consistently produced the lowest MAE and RMSE, excelling on PV and BESS as can be seen in Fig. 6, which can be attributed to its flexible non-parametric probabilistic design. Apart from DeepNPTS, N-BEATS achieved low errors on the PV datasets, demonstrating the effectiveness of deep residual MLP stacks. NLinear was consistently the weakest, failing to capture non-linear dynamics. All MLP-based models experienced increased errors with reduced sampling, while DeepNPTS and NHITS showed robustness under low data.

Fig. 7 shows that DeepAR achieved the best results among recurrent models, demonstrating robustness across smooth and volatile series through its probabilistic autoregressive structure. TFT followed closely, particularly on LCL 33 and ZHC datasets, due to its attention-based variable selection and gated residuals. MQ-RNN was inconsistent, occasionally competitive, but generated significantly high errors on the LCL dataset. All recurrent models showed reduced accuracy with lower sampling, with DeepAR and TFT showing the most resilience. In convolutional network-based models, TCN achieved the most accurate and robust forecasts as seen from Fig. 8, showing that its causal, dilated convolutions successfully captured long-range dependencies. TimesNet followed closely, benefiting from multi-frequency temporal

pattern extraction. TCN, BiTCN, and TimesNet remained resilient under reduced data, showing small error increases at 25% sampling rate. WaveNet consistently underperformed on PV and BESS, likely due to poor handling of irregularities. Similar to MQ-RNN, MQ-CNN produced highly varying results, showing poor performance on the LCL dataset. Performance generally declined with lower sampling rates for all models, but TCN and TimesNet were the most robust against data reduction.

Among transformer models, PatchTST was the most accurate across datasets and sampling rates as seen from Fig. 9. Its patch-based temporal encoding captured both local and long-range dependencies efficiently. iTransformer followed PatchTST, maintaining stable performance via adaptive attention. The Vanilla Transformer model generally performed poorly across all datasets. The accuracy declined with lower sampling for all, but PatchTST and iTransformer retained strong performance even at 25%. Fig. 10 shows the zero-shot performance of foundation models evaluated across multiple energy forecasting tasks and sampling rates. In general, TSFMs tend to produce forecasts with lower point-to-point variance, as seen in Figs. 11(f), 12(f), and 13(f), indicating that they do not effectively capture the short-term fluctuations present in the data during training. Timer-XL consistently achieved the lowest MAE and RMSE across most domains, particularly excelling on PV data, demonstrating strong generalization to diverse periodic patterns. TimesFM remained highly competitive, especially under reduced sampling rates. Across all models, accuracy declined as the sampling rate decreased. Timer-XL demonstrated the greatest overall stability and adaptability, confirming its suitability for both dense and sparse temporal data.

6.2. Diebold–Mariano tests

To determine whether differences in forecasting performance are statistically meaningful, we applied the Diebold–Mariano (DM) test to all pairwise model comparisons for each dataset and forecasting target (Diebold and Mariano, 2002; Hornek et al., 2025). The analysis used forecast versus actual time series from the 100% sampling rate outputs. When 10 runs were conducted for models with non-deterministic outputs, they were aligned in time and averaged so that each model contributed one representative forecast series per target. The DM procedure was carried out separately for two error perspectives using MAE and RMSE. For each model pair, we compared their per-timestamp loss values and evaluated whether one model consistently produced lower errors than the other. In our setup, the null hypothesis H_0 represents equal predictive accuracy between the two compared models, meaning there is no statistically significant evidence that one is better than the other. The alternative hypothesis H_1 suggests that the model on x-axis model is significantly more accurate than the one on the y-axis. The resulting p-values were organized into pairwise heat maps for all model combinations. The higher p-values ($p > 0.1$) indicate failure to reject H_0, i.e., no significant difference in predictive accuracy whereas lower p-values indicate evidence in favor of H_1, i.e., statistically significant superiority of the x-axis model relative to the y-axis model.

The DM test results across 3 forecasting tasks presented in Figs. 15–17 show several consistent and dataset-specific patterns in the statistical significance of forecast accuracy differences among the evaluated models. As observed from these heat maps, several pairwise comparisons yield high p-values, indicating that these models are not statistically distinguishable in terms of forecasting performance. For instance, TimeGPT and TimesFM exhibit no statistically significant difference in Fig. 15. However, a notable proportion of comparisons reveal statistically significant differences (e.g., $p \leq 0.05$), which are more frequently observed when contrasting naive models with other approaches. For example, LightGBM is shown to statistically outperform Naive Drift in Fig. 16. A noteworthy observation is the presence of several low p-values in the heat maps in Fig. 17, indicating that models such as MOIRAI, MQ-CNN and MQ-RNN consistently outperform the majority of the models. Overall, the DM test analysis highlights that while a number of forecasting models achieve similar levels of accuracy, meaningful statistical differences still emerge in specific cases, particularly when contrasting naive methods with advanced ML or DL approaches, as well as within certain datasets where model performance variability is more pronounced. The DM test results across the rest of the datasets are presented in Figs. A.28–A.37 in Appendix.

6.3. Overall outcome

The forecasting results show that LightGBM, DeepAR, PatchTST, DeepNPTS, TCN, and Timer-XL achieved the best performance within their respective model classes. These outcomes are based on the forecasting errors presented in Figs. 5–10, and are further supported by the DM tests shown in Figs. 15–17. The DM tests indicate that these models achieve statistically significantly better performance, on average, than the other models within their respective classes. It is important to note that the DM tests were conducted on a single test window per dataset (2–4 days), and that extending the evaluation to medium-term and long-term forecasting horizons may lead to different outcomes. Experimental results shown in Fig. 14 indicate that among the trained models, LightGBM consistently delivered the best performance, while DeepAR produced relatively the largest errors. The autoregressive recurrent design of DeepAR relies on dense temporal information and sequence structure, making it more vulnerable to recursive errors in comparison to other classes. These results demonstrate that, despite its relative simplicity, LightGBM serves as a strong baseline for short-term energy forecasting. It should be noted, however, that the performance rankings may change with model fine-tuning and may vary depending on the specific forecasting task. Nevertheless, LightGBM proved to be a suitable option for applications requiring reliable forecasts without extensive hyperparameter tuning, while maintaining low training and inference times, as shown in Table A.10 in Appendix. Moreover, the zero-shot forecasting results obtained with Timer-XL exhibit low errors, indicating that such TSFMs have strong potential for deployment in environments with limited training data.

7. Practical applications and limitations

Deployment in HEMS. Forecasting in HEMS often operates under constrained computational resources where models must deliver near real-time forecasts or even nowcasts. Our experiment results indicate that lightweight models such as LightGBM, the most robust ML model in our evaluation, can provide accurate forecasts with minimal errors, making it suitable for on-device deployment. More computationally intensive models such as PatchTST and Timer-XL can deliver higher accuracy for certain forecasting tasks but may require cloud-based execution or high-performance edge devices. A hybrid approach can therefore be a viable solution, for instance to have LightGBM or similar models for immediate nowcasts, and transformer or foundation models for batch forecasts in the cloud.

Deployment of TSFMs. TSFMs offer promising zero-shot forecasting capabilities; however, several practical limitations must be considered for real-world deployment. First, these models typically require substantial computational resources, including high memory usage and access to GPUs, which can limit their applicability in resource-constrained environments such as edge devices or embedded energy management systems. Second, inference latency can be higher compared to lightweight statistical or machine learning models, posing challenges for real-time or near real-time forecasting applications (Cai et al., 2026). Third, TSFMs are often pretrained on large and diverse datasets that may not fully capture domain-specific characteristics of energy consumption, potentially leading to suboptimal performance without fine-tuning (Karaouli et al., 2025). Additionally, data privacy and security concerns may also arise when relying on cloud-based deployment for large models. Finally, integrating TSFMs into existing

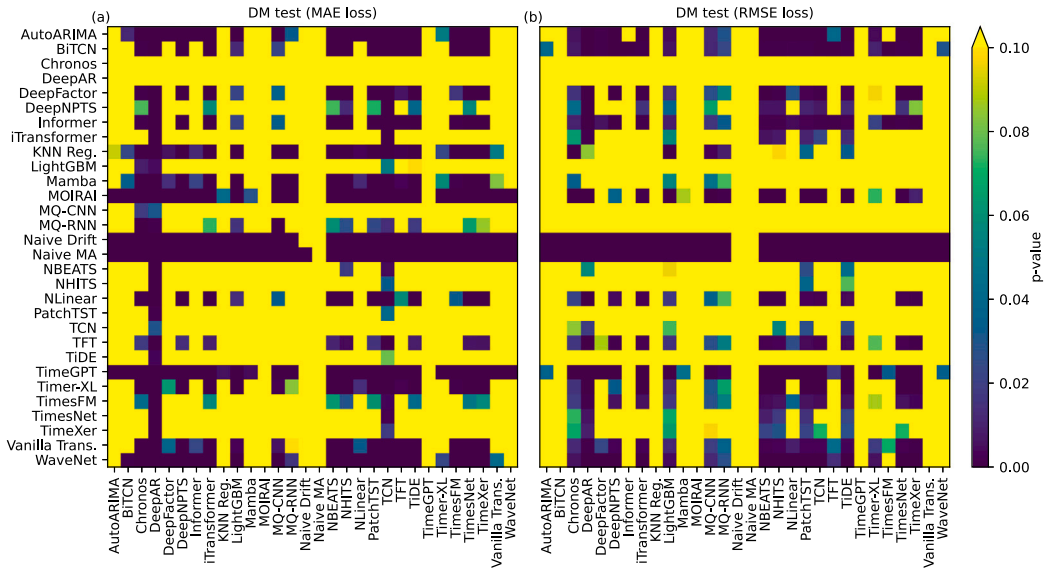


Fig. 15. DM statistical significance test for customer 9 in the ZHC dataset.

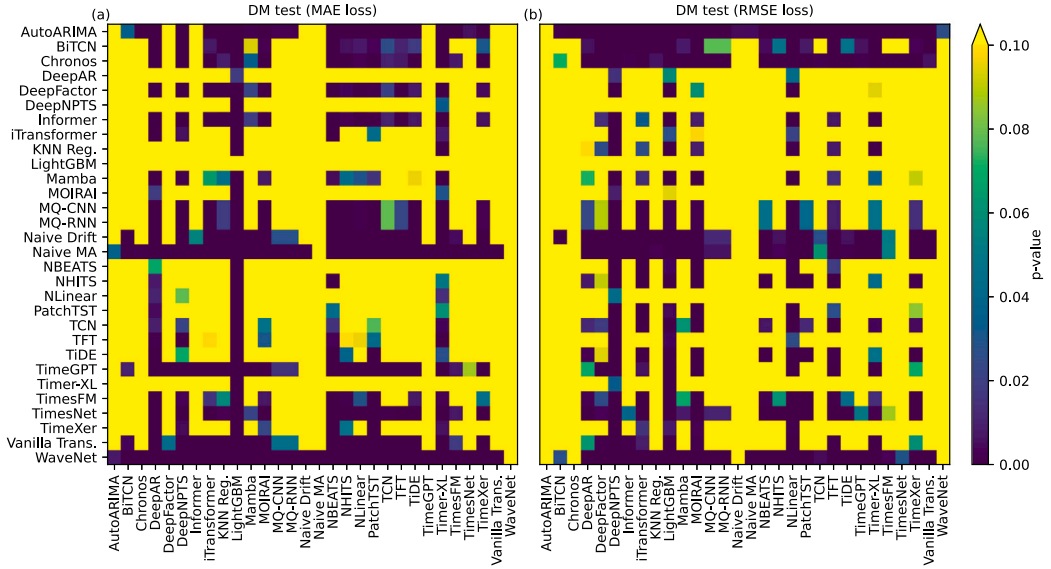


Fig. 16. DM statistical significance test for PV 1 in the BRD dataset.

operational pipelines can be complex due to dependencies on specific frameworks, hardware requirements, and limited standardization (Ferdaus et al., 2026).

Accuracy vs. Complexity. The trade-off between accuracy, interpretability, and computational burden is evident in our results. Simple statistical models such as AutoARIMA are interpretable but struggle with highly varying time series. LightGBM achieved the best accuracy among trained models, and since it is tree-based, it retains interpretability; hence, it is a practical alternative. DL models such as PatchTST, DeepNPTS, and TCN excelled in most forecasting tasks but at the expense of interpretability. In residential or regulatory contexts, in which explainability is critical, tree-based or interpretable neural models may be preferable over high-performing black box architectures.

Data Availability. Many residential HEMS scenarios involve limited historical data, which affects model viability. Our results show that models like LightGBM, TCN, and Timer-XL maintained strong performance even with a reduced sampling rate, demonstrating efficiency with low data availability. Foundation models such as TimesFM and Timer-XL can be fine-tuned to improve their performance, but since they are large models and require a large amount of data, they may not be favorable in applications in which data is scarce.

Privacy and Communication. Energy consumption data are privacy-sensitive, making centralized training problematic. While our study did not evaluate privacy-preserving methods directly, the computational and communication demands of DL models, such as Informer and iTransformer imply that federated learning and privacy-preserving data/model parameter sharing would be essential.

Computational Cost vs. Benefit. The performance of models such as PatchTST and Timer-XL without hyperparameter tuning suggests that, when fine-tuned, they potentially outperform LightGBM. However, this advantage comes at a substantial computational cost. In resource-constrained applications, such as residential households, these costs must be justified by clear operational gains to ensure they do not offset the benefits provided by energy and cost optimization techniques.

Hyperparameter Tuning. In this study, we used the default hyperparameters for all models to ensure a consistent and reproducible benchmarking framework and to evaluate off-the-shelf performance in applications with limited computational capacity. However, this

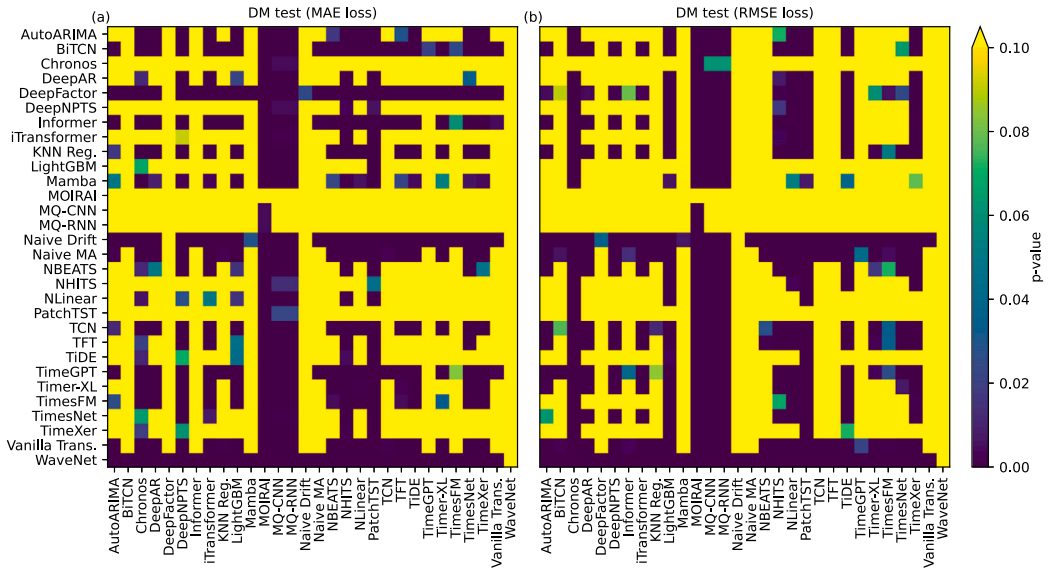


Fig. 17. DM statistical significance test for BESS 3 in the BRD dataset.

introduces a limitation, as hyperparameter optimization can significantly influence model performance, particularly for deep learning approaches. Consequently, the results presented in this work should be interpreted as baseline performance under standardized conditions rather than the absolute optimal capability of each model. In practical applications, where computational resources and domain expertise are available, targeted hyperparameter tuning is likely to produce different results. Therefore, future work should investigate systematic or automated hyperparameter optimization techniques to better capture the full capabilities of the evaluated models.

8. Conclusion

In this paper, we presented a comprehensive survey and benchmark of forecasting models, encompassing statistical, ML, DL, and foundation model categories. By evaluating 30 representative models across household load, PV generation, and BESS profiles using real-world datasets from 4 countries, we have provided a unified comparison under the same pre-processing, sampling rates, and evaluation metrics. Gradient boosting trees, represented by LightGBM, emerged as a dependable choice for accuracy and robustness among trained models, while TSFMs such as Timer-XL demonstrated that zero-shot capabilities can extend forecasting to data-scarce environments without the need for retraining the model. At the same time, the sensitivity of some models to reduced sampling underscores the importance of matching model architecture to data characteristics. These findings confirm that the choice of forecasting model should be guided by forecasting task characteristics, data availability, and operational constraints rather than by model accuracy alone. This study shows that foundation models can complement rather than replace established approaches, offering particular value in low-data scenarios. This work, therefore, provides both a benchmark and a strategic perspective for selecting and combining forecasting methods to meet the evolving needs of intelligent energy systems.

This study opens promising directions for future research. In particular, hybrid and ensemble approaches that combine the complementary strengths of statistical methods, ML, DL, and TSFMs could be explored to improve forecasting accuracy and robustness. For example, integrating the strong interpretability and stability of statistical models with the nonlinear representation learning capabilities of deep learning models, or combining zero-shot TSFM outputs with locally trained models, may yield superior performance across diverse datasets. Moreover, fine-tuning TSFMs on domain-specific energy data, as well as studying their deployment in real-world settings (e.g., edge versus cloud environments), represents an important direction for future work. Addressing challenges related to computational efficiency, interpretability, and scalability will be critical for utilizing these models in practical energy forecasting applications.

CRediT authorship contribution statement

Devilliers Dube: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Yu Xiang:** Writing – review & editing, Supervision. **Werner van Westering:** Writing – review & editing, Supervision. **Fedja Zdrnja:** Writing – review & editing, Supervision. **Koen Kok:** Writing – review & editing, Supervision. **Nirvana Meratnia:** Writing – review & editing, Supervision.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of the first draft, the first author used ChatGPT to improve the readability of the text. After using this tool, all authors reviewed and edited the content as needed, and, as such, they take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Extended experimental results

Tables A.4–A.9 present the comprehensive forecasting performance of all evaluated model categories across multiple electricity datasets, sampling rates, and error metrics. The results are reported using MAE and RMSE to provide a standardized basis for comparing forecasting accuracy. The results in Tables A.4 and A.9 correspond to single-run

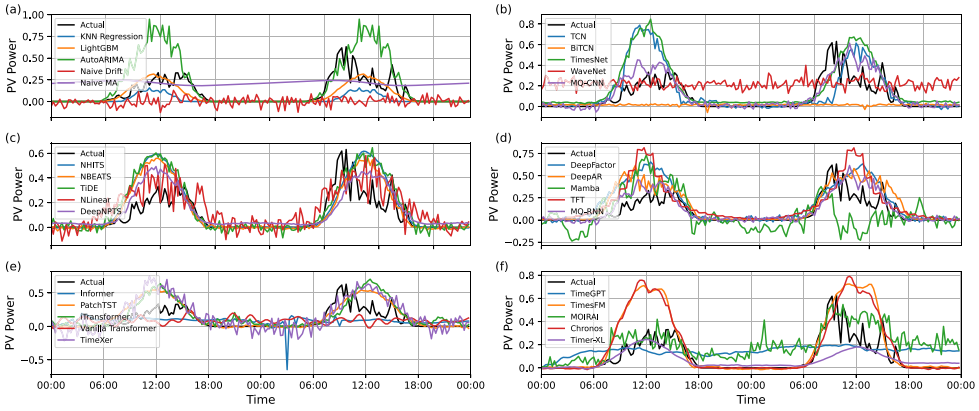


Fig. A.18. PV generation forecast results for house 2 in the BRD dataset.

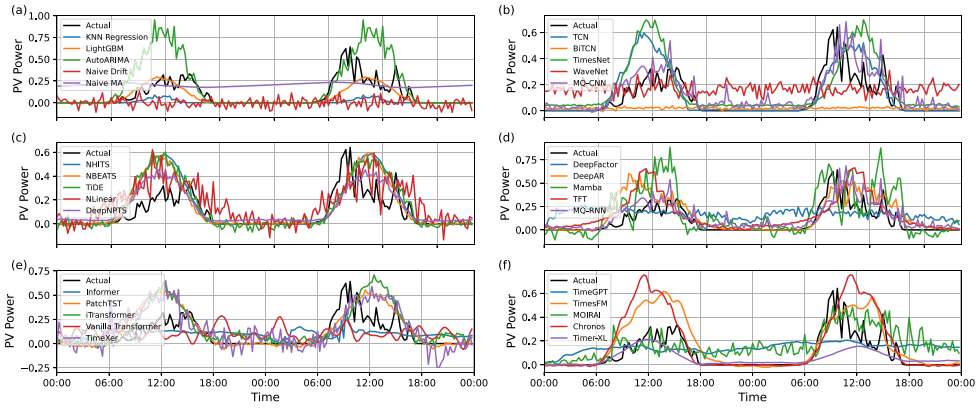


Fig. A.19. PV generation forecast results for house 3 in the BRD dataset.

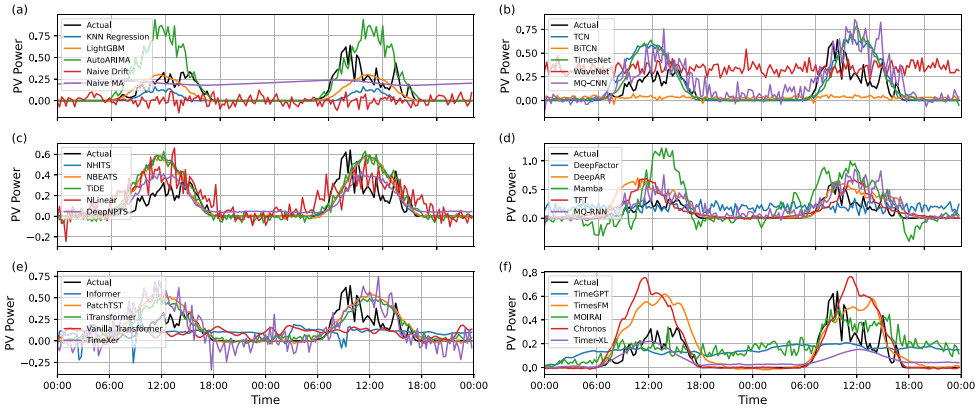


Fig. A.20. PV generation forecast results for house 4 in the BRD dataset.

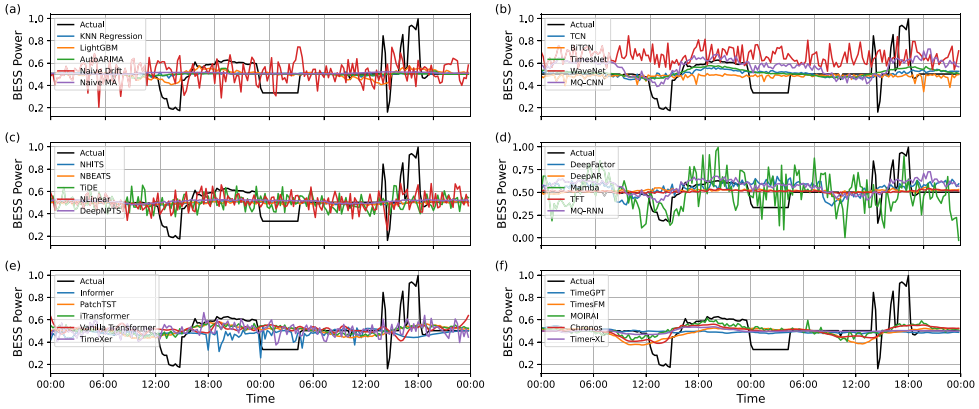


Fig. A.21. BESS operation forecast results for house 1 in the BRD dataset.

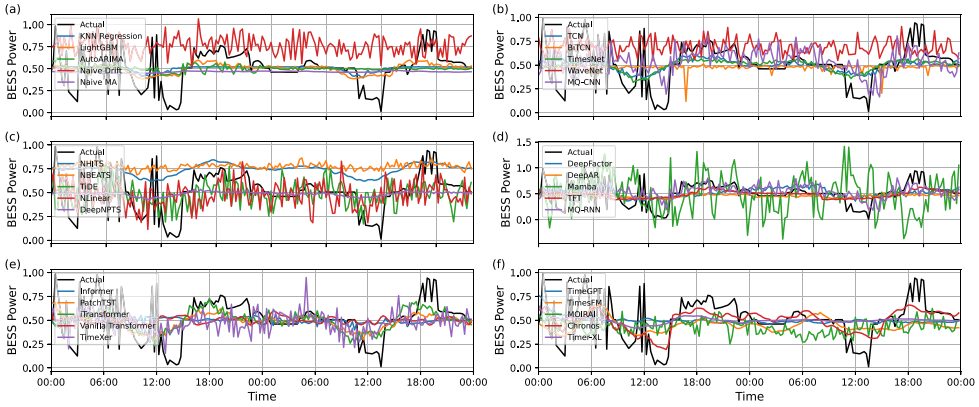


Fig. A.22. BESS operation forecast results for house 2 in the BRD dataset.

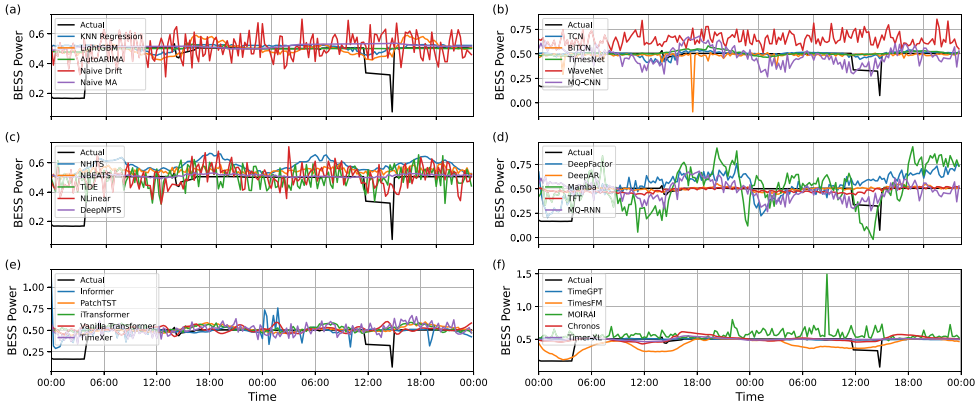


Fig. A.23. BESS operation forecast results for house 4 in the BRD dataset.

errors, as these models generate deterministic forecasts, whereas for all other models, the reported values represent the average performance over 10 independent runs for forecast horizons of $[192, 96, 64, 48]$ time steps produced by the respective sampling rates of $[100\%, 50\%, 33.33\%, 25\%]$. Collectively, these tables provide a complete and structured overview of the forecasting performance achieved by all models under varying data sampling conditions for the load, PV, and BESS time series datasets. Table A.10 shows the total training and inference times for the models across all datasets at different sampling rates. Figs. A.18–A.27 provide visual comparisons of the normalized actual versus forecast results at 100% sampling rate for all models. Figs. A.28–A.37 present the heat maps of the DM statistical significance tests, illustrating pairwise model comparisons across the remainder of the datasets.

Data availability

All the data used is publicly available.

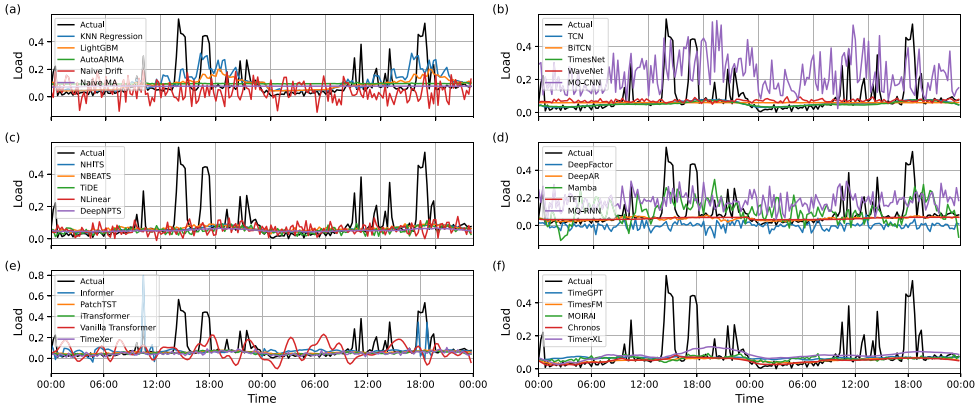


Fig. A.24. Load forecasting results for customer 8 in the ZHC dataset.

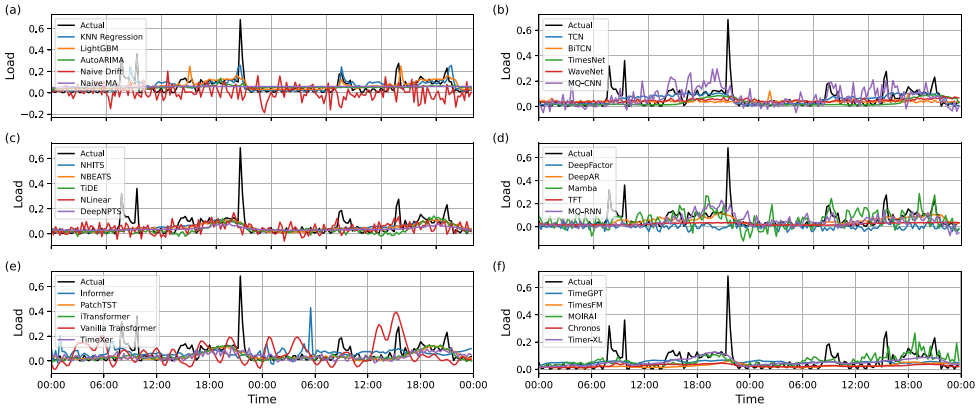


Fig. A.25. Load forecasting results for customer 43 in the ZHC dataset.

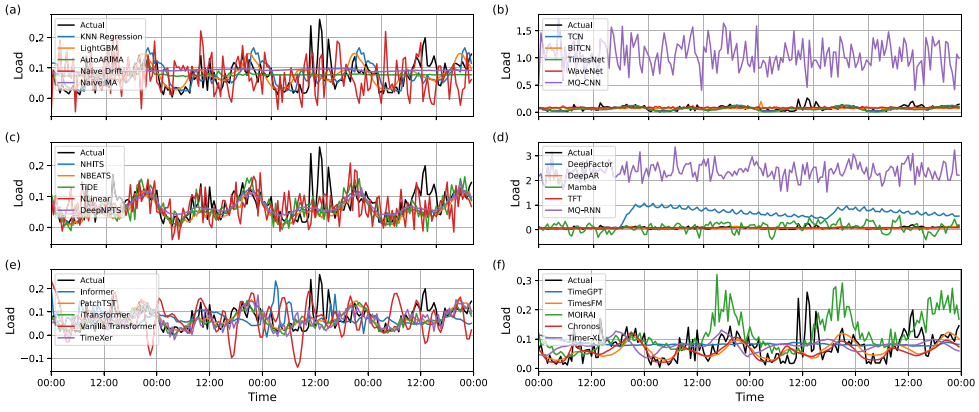


Fig. A.26. Load forecasting results for LCL 33 in the LCL dataset.

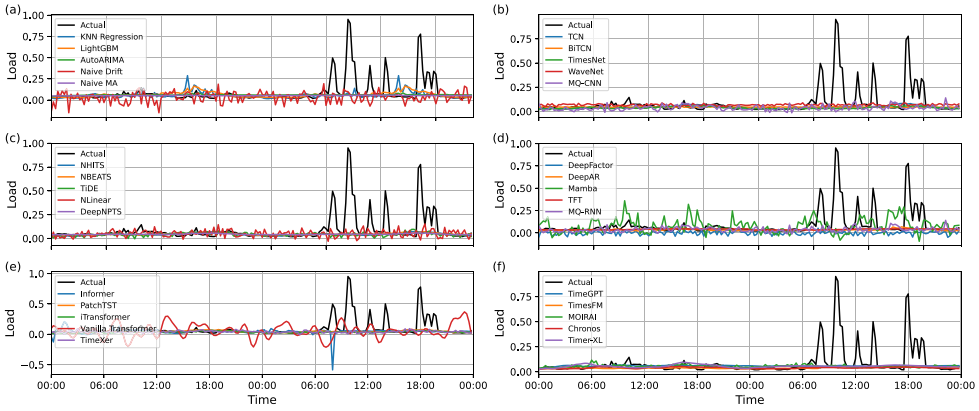


Fig. A.27. Load forecasting results for SFH 19 in the WPuQ dataset.

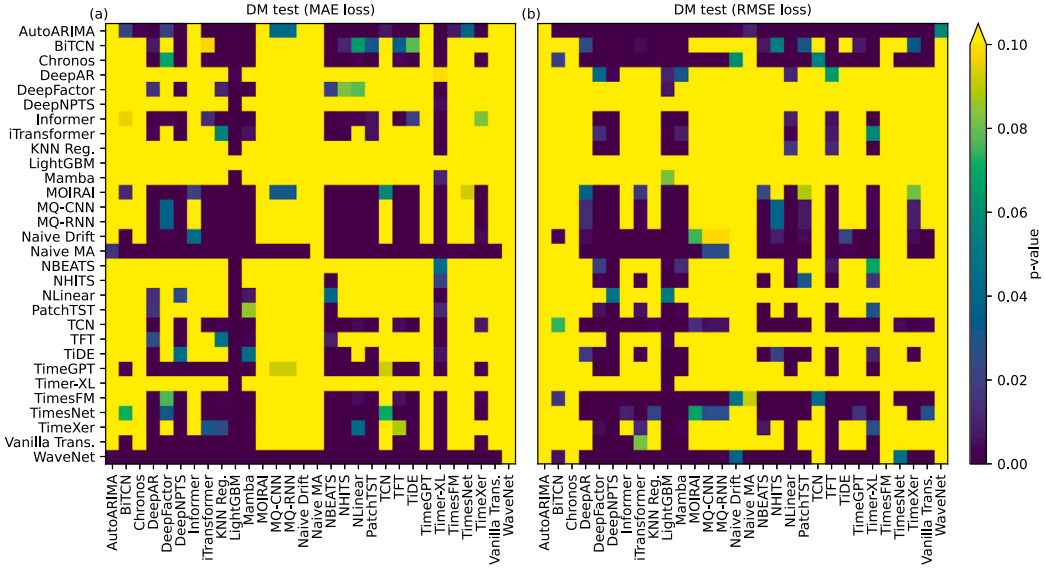


Fig. A.28. DM statistical significance test for PV 2 in the BRD dataset.

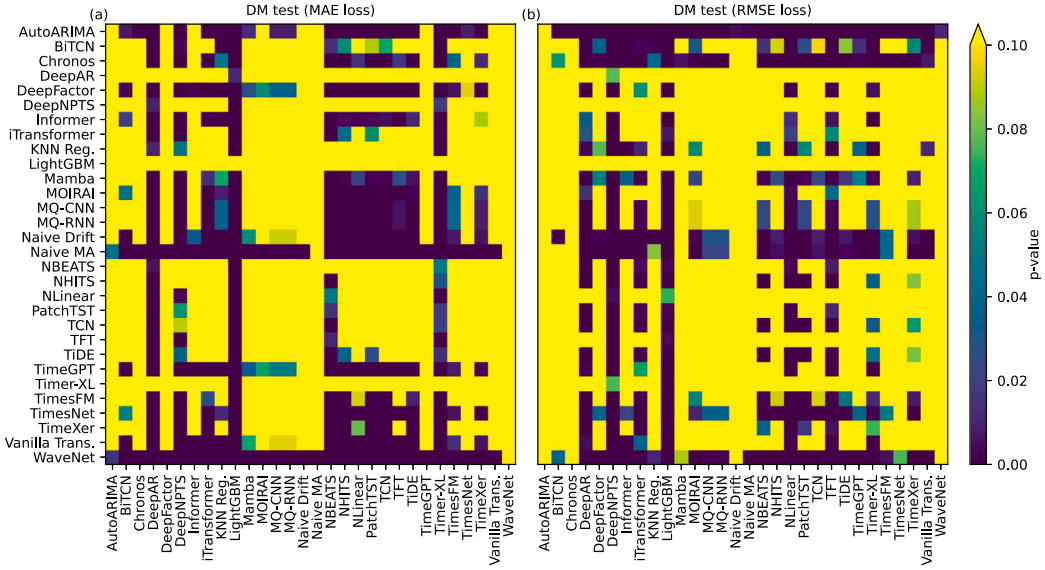


Fig. A.29. DM statistical significance test for PV 3 in the BRD dataset.

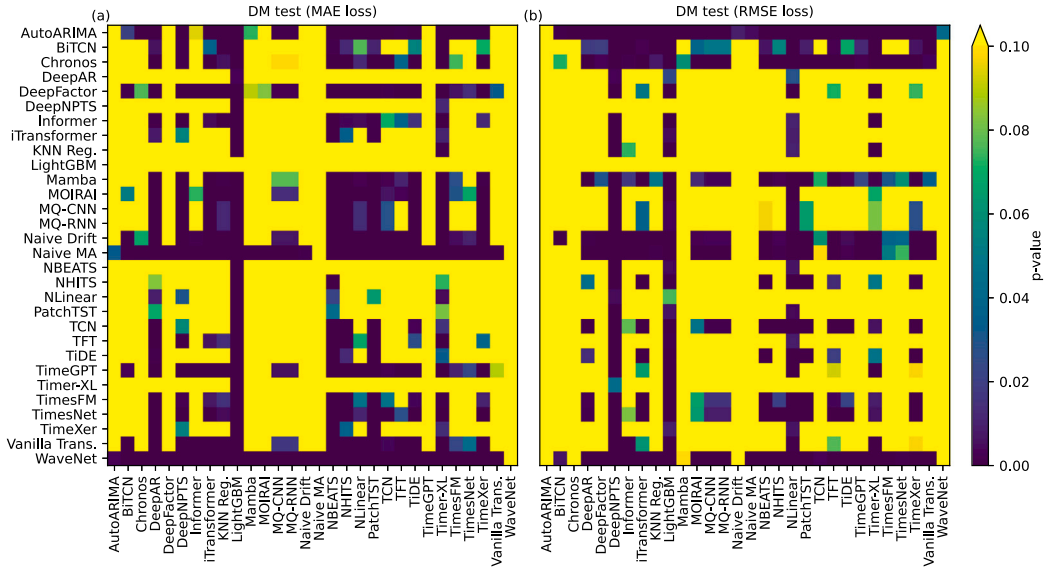


Fig. A.30. DM statistical significance test for PV 4 in the BRD dataset.

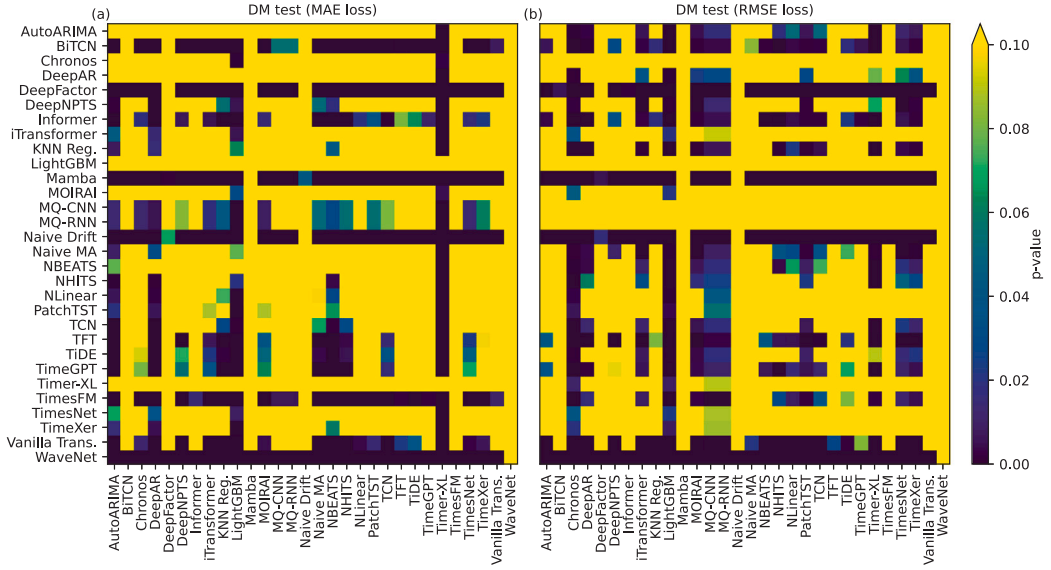


Fig. A.31. DM statistical significance test for BESS 1 in the BRD dataset.

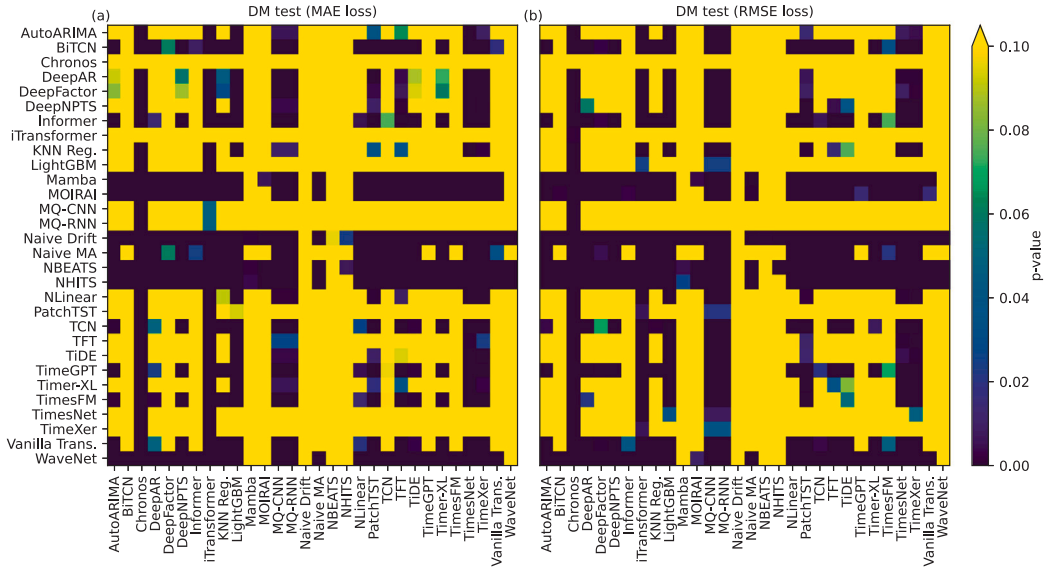


Fig. A.32. DM statistical significance test for BESS 2 in the BRD dataset.

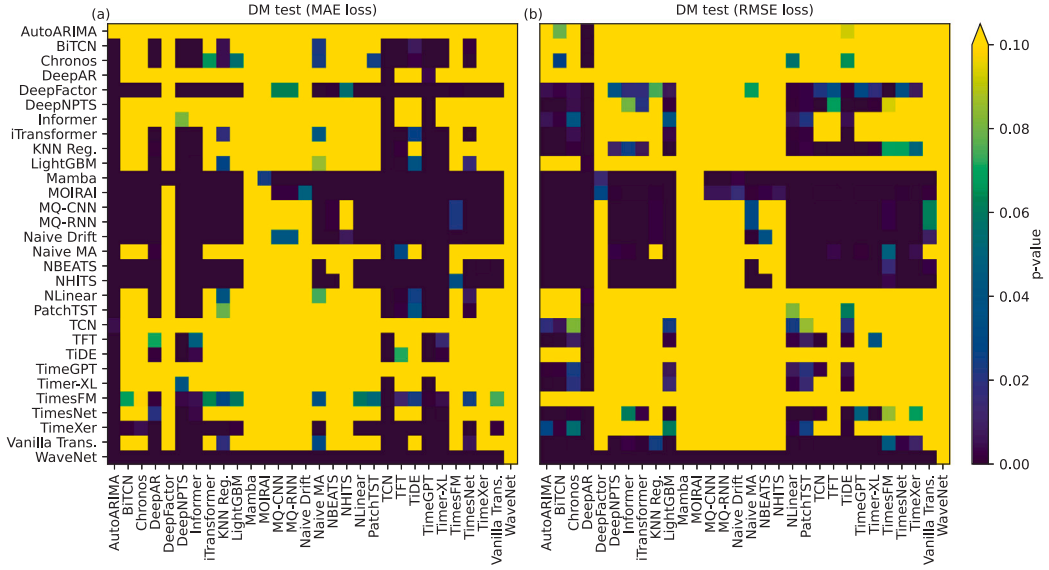


Fig. A.33. DM statistical significance test for BESS 4 in the BRD dataset.

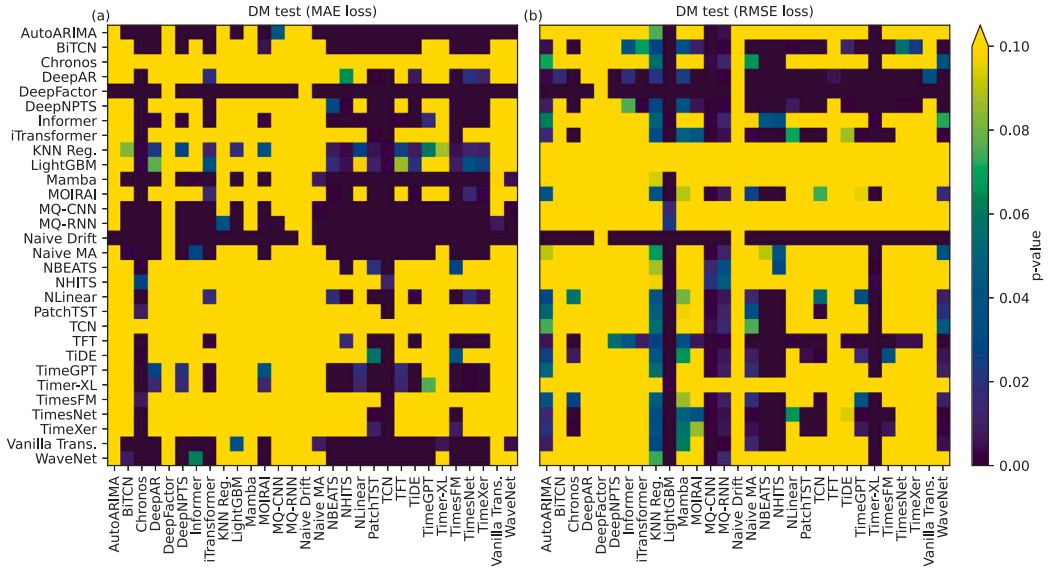


Fig. A.34. DM statistical significance test for customer 8 in the ZHC dataset.

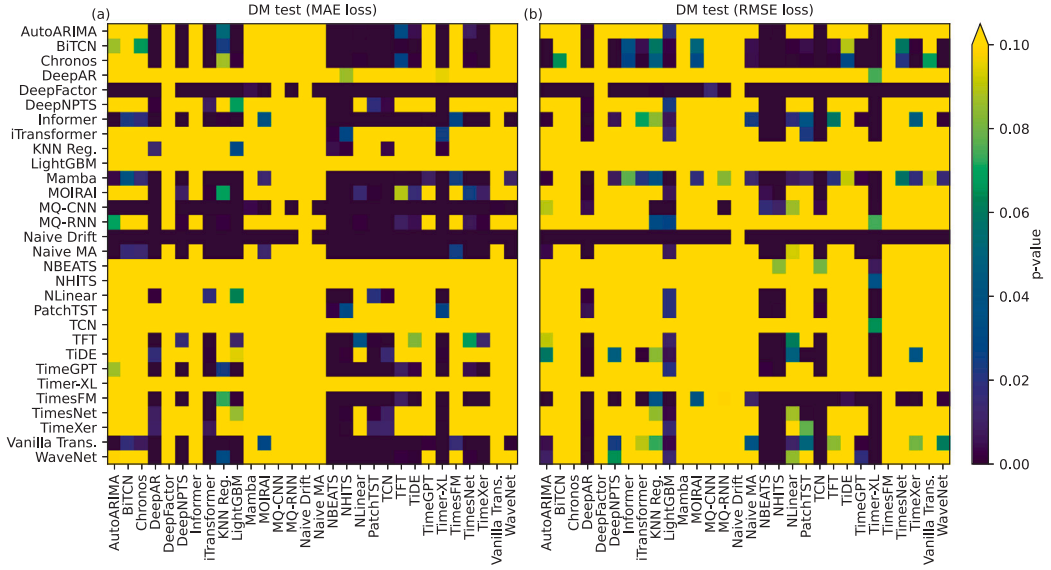


Fig. A.35. DM statistical significance test for customer 43 in the ZHC dataset.

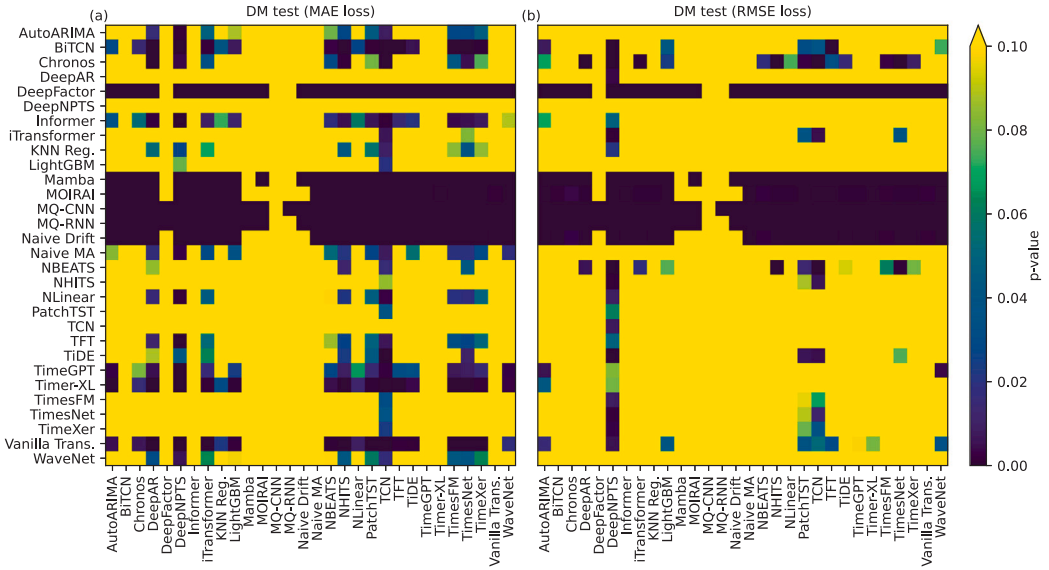


Fig. A.36. DM statistical significance test for LCL 33 in the LCL dataset.

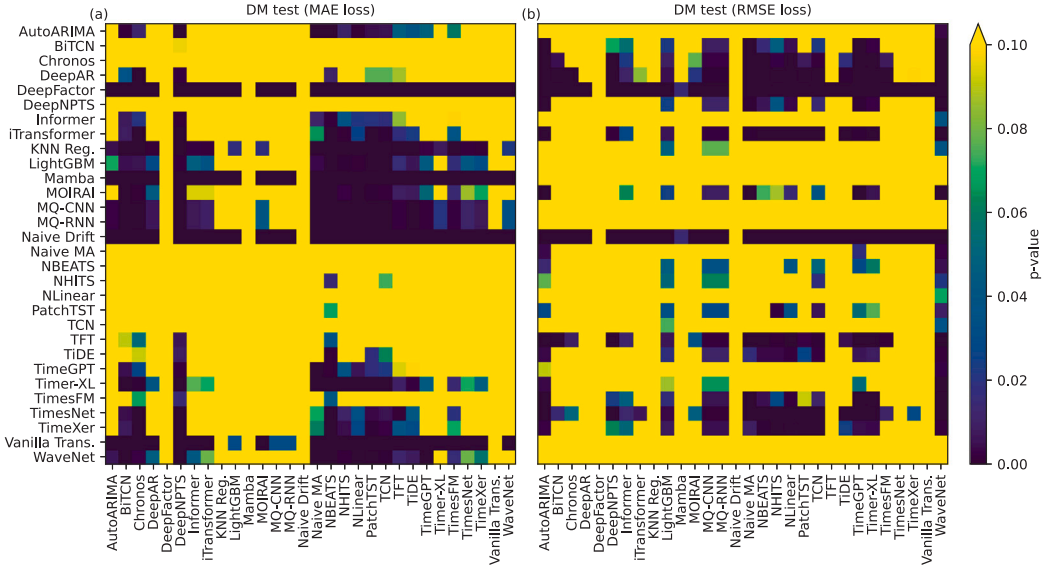


Fig. A.37. DM statistical significance test for SFH 19 in the WPuQ dataset.

Table A.4
Forecasting performance of statistical and conventional ML models across varying sampling rates, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
KNN Regression	100%	MAE	0.078	0.034	0.068	0.071	0.040	0.078	0.075	0.087	0.076	0.071	0.147	0.142	0.057
		RMSE	0.145	0.046	0.103	0.106	0.066	0.145	0.141	0.158	0.145	0.124	0.199	0.216	0.115
	50%	MAE	0.071	0.029	0.073	0.072	0.039	0.077	0.073	0.085	0.075	0.067	0.150	0.145	0.053
		RMSE	0.156	0.040	0.111	0.101	0.070	0.141	0.135	0.153	0.140	0.114	0.202	0.215	0.116
	33.33%	MAE	0.074	0.040	0.072	0.070	0.039	0.084	0.080	0.090	0.081	0.067	0.156	0.142	0.052
		RMSE	0.169	0.054	0.105	0.111	0.060	0.159	0.154	0.170	0.159	0.121	0.210	0.217	0.110
	25%	MAE	0.079	0.027	0.068	0.075	0.042	0.080	0.075	0.090	0.077	0.077	0.142	0.132	0.049
		RMSE	0.193	0.034	0.098	0.109	0.063	0.147	0.142	0.164	0.147	0.128	0.199	0.205	0.108
LightGBM	100%	MAE	0.064	0.030	0.062	0.050	0.036	0.050	0.049	0.049	0.050	0.068	0.136	0.125	0.061
		RMSE	0.146	0.042	0.102	0.089	0.066	0.100	0.099	0.102	0.102	0.111	0.181	0.180	0.105
	50%	MAE	0.062	0.030	0.065	0.048	0.036	0.050	0.048	0.049	0.050	0.065	0.133	0.125	0.065
		RMSE	0.146	0.038	0.106	0.079	0.073	0.096	0.094	0.097	0.097	0.104	0.173	0.178	0.108
	33.33%	MAE	0.072	0.032	0.069	0.044	0.040	0.058	0.056	0.057	0.058	0.067	0.145	0.127	0.060
		RMSE	0.168	0.043	0.104	0.085	0.064	0.115	0.112	0.116	0.115	0.112	0.191	0.183	0.101
	25%	MAE	0.078	0.027	0.056	0.053	0.035	0.048	0.046	0.047	0.048	0.067	0.124	0.119	0.063
		RMSE	0.186	0.035	0.090	0.089	0.055	0.099	0.097	0.100	0.101	0.109	0.168	0.176	0.103
AutoARIMA	100%	MAE	0.061	0.035	0.074	0.064	0.044	0.125	0.123	0.126	0.125	0.069	0.148	0.139	0.045
		RMSE	0.147	0.045	0.113	0.113	0.073	0.232	0.232	0.235	0.233	0.124	0.196	0.203	0.105
	50%	MAE	0.070	0.033	0.069	0.089	0.046	0.106	0.117	0.107	0.107	0.105	0.171	0.154	0.087
		RMSE	0.160	0.041	0.117	0.114	0.088	0.200	0.224	0.204	0.203	0.152	0.218	0.201	0.151
	33.33%	MAE	0.074	0.035	0.083	0.054	0.044	0.120	0.130	0.122	0.122	0.108	0.207	0.140	0.094
		RMSE	0.178	0.044	0.116	0.090	0.066	0.222	0.246	0.227	0.226	0.159	0.260	0.184	0.160
	25%	MAE	0.091	0.032	0.052	0.069	0.044	0.104	0.119	0.104	0.106	0.112	0.181	0.151	0.080
		RMSE	0.204	0.039	0.100	0.115	0.065	0.195	0.222	0.198	0.198	0.151	0.226	0.196	0.139
Naive Drift	100%	MAE	0.092	0.055	0.096	0.103	0.082	0.118	0.117	0.118	0.123	0.115	0.274	0.181	0.096
		RMSE	0.169	0.068	0.144	0.149	0.115	0.179	0.175	0.180	0.184	0.159	0.340	0.247	0.132
	50%	MAE	0.111	0.066	0.118	0.124	0.093	0.129	0.128	0.127	0.130	0.130	0.208	0.192	0.094
		RMSE	0.182	0.082	0.154	0.169	0.138	0.188	0.185	0.179	0.182	0.176	0.252	0.265	0.128
	33.33%	MAE	0.117	0.082	0.161	0.134	0.113	0.136	0.136	0.127	0.143	0.135	0.322	0.215	0.104
		RMSE	0.199	0.100	0.189	0.167	0.142	0.205	0.204	0.193	0.205	0.165	0.386	0.285	0.134
	25%	MAE	0.132	0.079	0.173	0.139	0.084	0.125	0.136	0.142	0.150	0.128	0.344	0.206	0.117
		RMSE	0.221	0.096	0.197	0.179	0.112	0.170	0.203	0.205	0.215	0.178	0.404	0.272	0.158
Naive MA	100%	MAE	0.059	0.039	0.061	0.083	0.046	0.161	0.161	0.161	0.163	0.069	0.162	0.148	0.052
		RMSE	0.149	0.049	0.119	0.120	0.081	0.180	0.179	0.179	0.181	0.125	0.213	0.218	0.107
	50%	MAE	0.059	0.037	0.062	0.080	0.049	0.160	0.159	0.160	0.163	0.069	0.157	0.149	0.052
		RMSE	0.148	0.046	0.122	0.113	0.093	0.178	0.176	0.177	0.179	0.119	0.212	0.224	0.111
	33.33%	MAE	0.068	0.036	0.069	0.088	0.046	0.174	0.174	0.174	0.177	0.067	0.164	0.147	0.043
		RMSE	0.174	0.046	0.126	0.115	0.076	0.195	0.194	0.194	0.196	0.122	0.219	0.221	0.106
	25%	MAE	0.079	0.044	0.055	0.085	0.048	0.168	0.167	0.168	0.169	0.073	0.159	0.142	0.045
		RMSE	0.188	0.051	0.099	0.118	0.076	0.182	0.182	0.182	0.183	0.122	0.210	0.221	0.105

Table A.5
Forecasting performance of MLP-based models across varying sampling rates, averaged over 10 runs, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
NHITS	100%	MAE	0.059	0.028	0.056	0.047	0.035	0.081	0.080	0.082	0.081	0.072	0.259	0.123	0.086
		RMSE	0.149	0.042	0.114	0.096	0.069	0.143	0.143	0.144	0.143	0.122	0.300	0.200	0.128
	50%	MAE	0.058	0.024	0.057	0.046	0.035	0.078	0.079	0.080	0.078	0.070	0.189	0.129	0.070
		RMSE	0.150	0.035	0.118	0.090	0.079	0.140	0.144	0.144	0.141	0.114	0.231	0.184	0.120
	33.33%	MAE	0.068	0.030	0.064	0.040	0.037	0.083	0.084	0.086	0.084	0.072	0.182	0.124	0.065
		RMSE	0.172	0.044	0.124	0.088	0.066	0.145	0.149	0.151	0.147	0.121	0.225	0.192	0.111
	25%	MAE	0.074	0.021	0.047	0.049	0.036	0.076	0.075	0.077	0.076	0.077	0.164	0.163	0.059
		RMSE	0.190	0.030	0.101	0.101	0.059	0.131	0.130	0.134	0.131	0.120	0.202	0.199	0.107
N-BEATS	100%	MAE	0.058	0.030	0.057	0.049	0.035	0.078	0.078	0.079	0.078	0.072	0.266	0.136	0.076
		RMSE	0.149	0.045	0.114	0.097	0.070	0.131	0.132	0.134	0.132	0.124	0.314	0.216	0.124
	50%	MAE	0.058	0.024	0.058	0.046	0.036	0.078	0.076	0.076	0.075	0.071	0.146	0.119	0.074
		RMSE	0.150	0.035	0.118	0.091	0.080	0.134	0.131	0.129	0.130	0.114	0.194	0.187	0.120
	33.33%	MAE	0.068	0.030	0.064	0.041	0.037	0.081	0.081	0.081	0.080	0.070	0.154	0.115	0.068
		RMSE	0.173	0.044	0.123	0.087	0.064	0.137	0.138	0.137	0.136	0.115	0.201	0.183	0.110
	25%	MAE	0.074	0.021	0.047	0.049	0.037	0.072	0.071	0.072	0.072	0.074	0.128	0.112	0.068
		RMSE	0.192	0.030	0.100	0.100	0.060	0.120	0.121	0.122	0.121	0.115	0.170	0.181	0.109
TiDE	100%	MAE	0.061	0.034	0.060	0.052	0.041	0.092	0.094	0.094	0.093	0.095	0.174	0.150	0.084
		RMSE	0.151	0.047	0.120	0.096	0.078	0.145	0.150	0.146	0.146	0.135	0.222	0.202	0.121
	50%	MAE	0.061	0.031	0.062	0.055	0.043	0.091	0.089	0.090	0.088	0.096	0.181	0.167	0.090
		RMSE	0.151	0.042	0.123	0.094	0.087	0.139	0.142	0.137	0.135	0.133	0.227	0.224	0.130
	33.33%	MAE	0.072	0.037	0.068	0.051	0.044	0.098	0.098	0.100	0.096	0.092	0.192	0.171	0.090
		RMSE	0.175	0.049	0.126	0.091	0.071	0.146	0.148	0.148	0.143	0.133	0.242	0.231	0.125
	25%	MAE	0.081	0.028	0.051	0.062	0.042	0.089	0.087	0.088	0.085	0.098	0.185	0.172	0.095
		RMSE	0.193	0.036	0.102	0.109	0.067	0.124	0.130	0.126	0.120	0.137	0.237	0.232	0.132
NLinear	100%	MAE	0.072	0.044	0.067	0.075	0.050	0.097	0.101	0.103	0.101	0.094	0.181	0.154	0.086
		RMSE	0.152	0.058	0.121	0.120	0.081	0.129	0.136	0.138	0.136	0.134	0.229	0.207	0.121
	50%	MAE	0.076	0.050	0.076	0.083	0.059	0.109	0.114	0.114	0.111	0.102	0.186	0.179	0.096
		RMSE	0.154	0.063	0.129	0.123	0.098	0.144	0.150	0.150	0.145	0.140	0.237	0.236	0.136
	33.33%	MAE	0.093	0.059	0.100	0.085	0.066	0.132	0.143	0.136	0.124	0.108	0.235	0.181	0.104
		RMSE	0.184	0.074	0.138	0.126	0.089	0.174	0.189	0.180	0.168	0.147	0.293	0.242	0.140
	25%	MAE	0.106	0.054	0.100	0.089	0.069	0.134	0.138	0.136	0.139	0.111	0.230	0.186	0.105
		RMSE	0.201	0.067	0.132	0.139	0.090	0.183	0.181	0.177	0.183	0.146	0.283	0.250	0.143
DeepNPTS	100%	MAE	0.058	0.029	0.058	0.050	0.040	0.075	0.077	0.075	0.078	0.073	0.149	0.126	0.051
		RMSE	0.150	0.041	0.120	0.103	0.075	0.101	0.107	0.105	0.104	0.123	0.200	0.201	0.113
	50%	MAE	0.057	0.027	0.060	0.050	0.041	0.075	0.075	0.077	0.075	0.072	0.139	0.131	0.050
		RMSE	0.150	0.036	0.124	0.097	0.085	0.099	0.102	0.101	0.101	0.116	0.189	0.208	0.114
	33.33%	MAE	0.067	0.032	0.066	0.044	0.043	0.085	0.087	0.088	0.084	0.069	0.161	0.128	0.056
		RMSE	0.173	0.044	0.126	0.095	0.072	0.120	0.122	0.120	0.117	0.120	0.214	0.206	0.110
	25%	MAE	0.073	0.023	0.048	0.054	0.040	0.071	0.073	0.072	0.071	0.075	0.137	0.125	0.051
		RMSE	0.190	0.030	0.102	0.107	0.066	0.100	0.102	0.102	0.102	0.119	0.189	0.207	0.108

Table A.6
Forecasting performance of recurrent network-based models across varying sampling rates, averaged over 10 runs, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
DeepFactor	100%	MAE	0.091	0.450	0.096	0.068	0.065	0.123	0.120	0.127	0.137	0.109	0.160	0.204	0.107
		RMSE	0.171	0.513	0.149	0.115	0.100	0.151	0.159	0.150	0.163	0.147	0.200	0.232	0.140
	50%	MAE	0.091	0.421	0.099	0.058	0.069	0.148	0.132	0.162	0.149	0.104	0.138	0.178	0.107
		RMSE	0.172	0.478	0.153	0.103	0.110	0.230	0.177	0.214	0.208	0.139	0.172	0.205	0.142
	33.33%	MAE	0.100	0.209	0.105	0.069	0.062	0.141	0.162	0.145	0.154	0.103	0.172	0.123	0.124
		RMSE	0.195	0.239	0.158	0.110	0.092	0.240	0.239	0.242	0.223	0.139	0.211	0.155	0.151
	25%	MAE	0.108	0.224	0.079	0.067	0.068	0.143	0.135	0.164	0.173	0.104	0.164	0.161	0.154
		RMSE	0.215	0.255	0.126	0.111	0.094	0.221	0.210	0.246	0.252	0.136	0.197	0.188	0.174
DeepAR	100%	MAE	0.059	0.029	0.058	0.044	0.036	0.105	0.099	0.099	0.101	0.069	0.156	0.144	0.046
		RMSE	0.149	0.043	0.122	0.095	0.069	0.161	0.158	0.156	0.160	0.125	0.211	0.219	0.113
	50%	MAE	0.058	0.026	0.057	0.042	0.037	0.103	0.099	0.098	0.099	0.067	0.157	0.148	0.049
		RMSE	0.149	0.036	0.123	0.088	0.080	0.157	0.153	0.155	0.153	0.119	0.212	0.224	0.117
	33.33%	MAE	0.069	0.028	0.063	0.041	0.038	0.111	0.107	0.106	0.108	0.065	0.162	0.145	0.045
		RMSE	0.172	0.041	0.124	0.087	0.065	0.176	0.172	0.173	0.173	0.122	0.219	0.222	0.107
	25%	MAE	0.075	0.021	0.044	0.045	0.042	0.106	0.101	0.100	0.102	0.071	0.148	0.142	0.045
		RMSE	0.191	0.030	0.099	0.094	0.066	0.165	0.161	0.162	0.162	0.122	0.207	0.221	0.109
Mamba	100%	MAE	0.105	0.138	0.083	0.117	0.067	0.143	0.140	0.163	0.165	0.172	0.346	0.187	0.156
		RMSE	0.162	0.175	0.120	0.157	0.097	0.204	0.192	0.233	0.247	0.218	0.426	0.243	0.193
	50%	MAE	0.110	0.104	0.081	0.135	0.065	0.158	0.290	0.283	0.217	0.167	0.415	0.174	0.140
		RMSE	0.166	0.134	0.119	0.183	0.096	0.227	0.420	0.428	0.304	0.216	0.522	0.222	0.176
	33.33%	MAE	0.111	0.078	0.078	0.123	0.055	0.275	0.319	0.379	0.495	0.170	0.513	0.145	0.172
		RMSE	0.177	0.100	0.115	0.170	0.077	0.395	0.470	0.586	0.716	0.221	0.626	0.189	0.217
	25%	MAE	0.129	0.049	0.070	0.119	0.060	0.262	0.187	0.320	0.244	0.189	0.379	0.166	0.171
		RMSE	0.211	0.062	0.101	0.159	0.083	0.375	0.269	0.481	0.355	0.239	0.472	0.213	0.215
TFT	100%	MAE	0.059	0.036	0.058	0.055	0.043	0.094	0.100	0.098	0.100	0.077	0.146	0.137	0.057
		RMSE	0.153	0.046	0.121	0.112	0.076	0.138	0.147	0.142	0.149	0.127	0.197	0.218	0.112
	50%	MAE	0.058	0.033	0.059	0.053	0.044	0.089	0.102	0.089	0.091	0.071	0.147	0.151	0.058
		RMSE	0.153	0.041	0.123	0.105	0.086	0.135	0.161	0.131	0.137	0.118	0.197	0.235	0.113
	33.33%	MAE	0.069	0.036	0.065	0.050	0.043	0.104	0.102	0.107	0.101	0.071	0.159	0.149	0.061
		RMSE	0.178	0.047	0.125	0.100	0.070	0.163	0.153	0.153	0.151	0.122	0.207	0.227	0.111
	25%	MAE	0.074	0.030	0.047	0.056	0.046	0.081	0.098	0.089	0.099	0.074	0.141	0.143	0.055
		RMSE	0.195	0.036	0.100	0.113	0.070	0.123	0.136	0.138	0.145	0.122	0.197	0.223	0.106
MQ-RNN	100%	MAE	0.076	0.976	0.087	0.064	0.061	0.108	0.137	0.113	0.108	0.083	0.155	0.127	0.093
		RMSE	0.152	1.008	0.127	0.103	0.087	0.158	0.188	0.163	0.155	0.120	0.198	0.172	0.132
	50%	MAE	0.069	0.620	0.069	0.070	0.071	0.112	0.128	0.088	0.100	0.082	0.153	0.136	0.095
		RMSE	0.152	0.645	0.118	0.094	0.104	0.153	0.167	0.134	0.152	0.120	0.191	0.186	0.137
	33.33%	MAE	0.108	0.774	0.084	0.058	0.057	0.100	0.123	0.094	0.116	0.091	0.158	0.126	0.113
		RMSE	0.200	0.801	0.124	0.097	0.079	0.149	0.174	0.136	0.164	0.127	0.199	0.180	0.148
	25%	MAE	0.082	0.214	0.059	0.068	0.052	0.075	0.097	0.088	0.081	0.085	0.147	0.116	0.087
		RMSE	0.193	0.245	0.104	0.098	0.078	0.114	0.139	0.134	0.124	0.123	0.186	0.171	0.123

Table A.7
Forecasting performance of convolutional network-based models across varying sampling rates, averaged over 10 runs, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
TCN	100%	MAE	0.058	0.027	0.054	0.046	0.036	0.098	0.117	0.084	0.091	0.074	0.160	0.144	0.049
		RMSE	0.148	0.041	0.117	0.097	0.069	0.172	0.196	0.148	0.162	0.123	0.208	0.217	0.110
	50%	MAE	0.056	0.024	0.056	0.043	0.038	0.090	0.102	0.086	0.080	0.070	0.146	0.148	0.050
		RMSE	0.148	0.034	0.120	0.093	0.081	0.158	0.173	0.153	0.142	0.111	0.190	0.222	0.114
	33.33%	MAE	0.067	0.029	0.062	0.040	0.040	0.090	0.096	0.103	0.094	0.064	0.166	0.141	0.048
		RMSE	0.173	0.043	0.121	0.091	0.066	0.155	0.161	0.182	0.167	0.116	0.219	0.218	0.105
	25%	MAE	0.073	0.021	0.044	0.048	0.038	0.082	0.073	0.085	0.082	0.069	0.139	0.141	0.044
		RMSE	0.190	0.030	0.095	0.102	0.060	0.145	0.127	0.147	0.139	0.112	0.181	0.219	0.107
BiTCN	100%	MAE	0.059	0.038	0.062	0.060	0.047	0.103	0.099	0.098	0.100	0.086	0.167	0.159	0.064
		RMSE	0.151	0.049	0.121	0.119	0.080	0.167	0.166	0.165	0.166	0.135	0.218	0.217	0.112
	50%	MAE	0.058	0.036	0.062	0.057	0.049	0.102	0.098	0.093	0.098	0.081	0.166	0.167	0.064
		RMSE	0.152	0.043	0.123	0.110	0.091	0.165	0.158	0.154	0.163	0.127	0.216	0.223	0.114
	33.33%	MAE	0.068	0.038	0.070	0.052	0.047	0.108	0.105	0.104	0.105	0.080	0.170	0.159	0.062
		RMSE	0.176	0.048	0.126	0.107	0.075	0.179	0.180	0.177	0.181	0.132	0.224	0.219	0.104
	25%	MAE	0.074	0.029	0.050	0.059	0.048	0.101	0.099	0.098	0.100	0.088	0.157	0.154	0.052
		RMSE	0.194	0.035	0.102	0.116	0.071	0.168	0.169	0.169	0.169	0.131	0.210	0.218	0.105
TimesNet	100%	MAE	0.060	0.029	0.056	0.049	0.041	0.108	0.112	0.109	0.103	0.073	0.139	0.130	0.058
		RMSE	0.153	0.043	0.119	0.099	0.076	0.167	0.175	0.168	0.161	0.118	0.187	0.208	0.113
	50%	MAE	0.058	0.025	0.059	0.049	0.039	0.098	0.105	0.101	0.102	0.070	0.134	0.134	0.053
		RMSE	0.153	0.036	0.123	0.096	0.083	0.152	0.170	0.156	0.163	0.114	0.178	0.213	0.116
	33.33%	MAE	0.070	0.031	0.066	0.044	0.045	0.109	0.104	0.101	0.105	0.068	0.155	0.134	0.057
		RMSE	0.177	0.045	0.127	0.091	0.074	0.170	0.170	0.158	0.167	0.117	0.207	0.213	0.110
	25%	MAE	0.075	0.022	0.048	0.053	0.039	0.091	0.099	0.092	0.088	0.073	0.142	0.129	0.055
		RMSE	0.194	0.031	0.102	0.107	0.065	0.145	0.157	0.146	0.136	0.117	0.190	0.208	0.109
WaveNet	100%	MAE	0.063	0.038	0.066	0.066	0.046	0.173	0.183	0.173	0.184	0.234	0.216	0.298	0.288
		RMSE	0.146	0.048	0.116	0.115	0.077	0.194	0.204	0.195	0.206	0.265	0.277	0.368	0.318
	50%	MAE	0.064	0.038	0.069	0.063	0.048	0.189	0.176	0.193	0.186	0.224	0.225	0.303	0.314
		RMSE	0.147	0.047	0.118	0.108	0.087	0.211	0.197	0.215	0.207	0.255	0.286	0.373	0.346
	33.33%	MAE	0.070	0.036	0.069	0.063	0.047	0.178	0.180	0.174	0.183	0.227	0.215	0.271	0.342
		RMSE	0.171	0.046	0.121	0.105	0.072	0.200	0.203	0.196	0.205	0.257	0.273	0.347	0.370
	25%	MAE	0.084	0.034	0.058	0.067	0.051	0.177	0.178	0.194	0.183	0.236	0.205	0.232	0.271
		RMSE	0.186	0.041	0.100	0.115	0.066	0.197	0.198	0.214	0.204	0.265	0.266	0.308	0.303
MQ-CNN	100%	MAE	0.076	1.366	0.084	0.063	0.080	0.108	0.137	0.113	0.108	0.083	0.155	0.127	0.093
		RMSE	0.152	1.396	0.127	0.100	0.107	0.158	0.188	0.163	0.155	0.120	0.198	0.172	0.132
	50%	MAE	0.075	0.965	0.087	0.061	0.094	0.092	0.112	0.105	0.112	0.088	0.146	0.112	0.095
		RMSE	0.150	0.987	0.124	0.084	0.119	0.138	0.166	0.152	0.163	0.121	0.185	0.161	0.132
	33.33%	MAE	0.100	0.674	0.089	0.070	0.095	0.123	0.113	0.107	0.110	0.082	0.169	0.113	0.095
		RMSE	0.182	0.694	0.123	0.104	0.113	0.171	0.159	0.159	0.160	0.119	0.210	0.165	0.131
	25%	MAE	0.084	0.442	0.067	0.065	0.053	0.091	0.117	0.099	0.088	0.085	0.150	0.108	0.097
		RMSE	0.191	0.472	0.106	0.092	0.076	0.142	0.163	0.147	0.139	0.120	0.187	0.155	0.138

Table A.8
Forecasting performance of transformer-based models across varying sampling rates, averaged over 10 runs, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
Informer	100%	MAE	0.069	0.045	0.069	0.062	0.053	0.112	0.112	0.111	0.111	0.082	0.165	0.161	0.070
		RMSE	0.154	0.063	0.124	0.119	0.084	0.147	0.145	0.147	0.145	0.131	0.216	0.219	0.125
	50%	MAE	0.069	0.044	0.074	0.061	0.055	0.113	0.113	0.114	0.110	0.088	0.171	0.163	0.070
		RMSE	0.159	0.059	0.128	0.112	0.093	0.146	0.142	0.149	0.140	0.133	0.221	0.223	0.126
	33.33%	MAE	0.081	0.046	0.082	0.059	0.058	0.117	0.125	0.118	0.116	0.083	0.171	0.168	0.065
		RMSE	0.180	0.067	0.132	0.110	0.086	0.155	0.173	0.160	0.154	0.130	0.224	0.226	0.112
	25%	MAE	0.085	0.036	0.064	0.067	0.059	0.112	0.118	0.110	0.117	0.089	0.160	0.165	0.065
		RMSE	0.194	0.046	0.106	0.120	0.081	0.151	0.157	0.140	0.152	0.131	0.213	0.225	0.113
PatchTST	100%	MAE	0.058	0.029	0.055	0.048	0.038	0.081	0.085	0.085	0.081	0.074	0.143	0.119	0.061
		RMSE	0.150	0.042	0.117	0.093	0.074	0.132	0.139	0.136	0.131	0.118	0.186	0.180	0.105
	50%	MAE	0.057	0.025	0.057	0.047	0.038	0.084	0.087	0.083	0.084	0.070	0.130	0.122	0.057
		RMSE	0.149	0.035	0.120	0.089	0.083	0.136	0.143	0.134	0.137	0.112	0.167	0.187	0.108
	33.33%	MAE	0.067	0.030	0.066	0.043	0.040	0.094	0.094	0.093	0.093	0.066	0.155	0.124	0.054
		RMSE	0.172	0.043	0.121	0.085	0.070	0.149	0.152	0.148	0.147	0.116	0.202	0.193	0.102
	25%	MAE	0.074	0.023	0.051	0.053	0.037	0.086	0.090	0.086	0.086	0.071	0.136	0.121	0.053
		RMSE	0.189	0.030	0.097	0.101	0.064	0.135	0.145	0.136	0.135	0.116	0.173	0.197	0.102
iTransformer	100%	MAE	0.061	0.030	0.057	0.052	0.039	0.087	0.093	0.090	0.091	0.074	0.131	0.131	0.065
		RMSE	0.152	0.044	0.119	0.101	0.076	0.125	0.136	0.134	0.135	0.119	0.179	0.203	0.113
	50%	MAE	0.060	0.027	0.059	0.050	0.044	0.088	0.095	0.091	0.093	0.073	0.132	0.132	0.062
		RMSE	0.153	0.037	0.123	0.092	0.088	0.128	0.142	0.132	0.135	0.112	0.175	0.204	0.113
	33.33%	MAE	0.071	0.032	0.065	0.046	0.044	0.099	0.103	0.093	0.095	0.071	0.152	0.127	0.066
		RMSE	0.176	0.047	0.126	0.089	0.073	0.147	0.150	0.135	0.140	0.116	0.200	0.192	0.107
	25%	MAE	0.076	0.024	0.046	0.054	0.041	0.085	0.090	0.087	0.093	0.075	0.127	0.133	0.069
		RMSE	0.193	0.032	0.100	0.104	0.068	0.125	0.129	0.125	0.132	0.114	0.171	0.208	0.110
Vanilla Transformer	100%	MAE	0.100	0.063	0.088	0.090	0.069	0.125	0.126	0.125	0.125	0.098	0.172	0.172	0.085
		RMSE	0.168	0.080	0.132	0.139	0.100	0.161	0.160	0.161	0.162	0.142	0.224	0.239	0.136
	50%	MAE	0.091	0.054	0.090	0.079	0.075	0.118	0.121	0.115	0.123	0.088	0.170	0.168	0.091
		RMSE	0.162	0.066	0.139	0.123	0.113	0.153	0.154	0.150	0.159	0.129	0.220	0.236	0.139
	33.33%	MAE	0.104	0.056	0.088	0.075	0.066	0.123	0.121	0.126	0.128	0.086	0.171	0.159	0.073
		RMSE	0.186	0.071	0.132	0.119	0.090	0.164	0.160	0.166	0.168	0.133	0.225	0.231	0.120
	25%	MAE	0.098	0.046	0.067	0.071	0.065	0.119	0.126	0.115	0.128	0.092	0.156	0.150	0.073
		RMSE	0.196	0.058	0.108	0.122	0.084	0.153	0.159	0.150	0.164	0.131	0.210	0.227	0.118
TimeXer	100%	MAE	0.064	0.035	0.058	0.060	0.043	0.118	0.124	0.123	0.115	0.086	0.154	0.150	0.085
		RMSE	0.153	0.049	0.120	0.106	0.077	0.156	0.165	0.165	0.155	0.126	0.199	0.206	0.122
	50%	MAE	0.061	0.028	0.060	0.053	0.043	0.104	0.108	0.097	0.103	0.077	0.141	0.139	0.070
		RMSE	0.153	0.038	0.123	0.097	0.087	0.149	0.149	0.134	0.147	0.116	0.182	0.201	0.111
	33.33%	MAE	0.071	0.032	0.067	0.046	0.044	0.104	0.104	0.104	0.101	0.073	0.154	0.133	0.073
		RMSE	0.175	0.045	0.126	0.091	0.073	0.148	0.150	0.146	0.140	0.116	0.203	0.195	0.109
	25%	MAE	0.078	0.024	0.047	0.054	0.041	0.088	0.093	0.093	0.094	0.078	0.128	0.132	0.076
		RMSE	0.193	0.033	0.101	0.105	0.068	0.121	0.131	0.126	0.128	0.115	0.167	0.201	0.112

Table A.9
Zero-shot forecasting performance of time series foundation models across varying sampling rates, evaluated using MAE and RMSE metrics on multiple electricity datasets.

Model	Sampling	Metric	SFH 19	LCL 33	ZHC 8	ZHC 9	ZHC 43	PV 1	PV 2	PV 3	PV 4	BESS 1	BESS 2	BESS 3	BESS 4
TimeGPT	100%	MAE	0.060	0.039	0.060	0.084	0.045	0.122	0.123	0.122	0.123	0.076	0.163	0.156	0.048
		RMSE	0.147	0.049	0.117	0.118	0.074	0.142	0.141	0.141	0.142	0.126	0.215	0.218	0.111
	50%	MAE	0.062	0.039	0.062	0.065	0.043	0.119	0.120	0.118	0.119	0.084	0.148	0.141	0.082
		RMSE	0.148	0.054	0.121	0.102	0.083	0.139	0.140	0.138	0.139	0.124	0.197	0.191	0.117
	33.33%	MAE	0.070	0.043	0.074	0.076	0.045	0.170	0.169	0.170	0.171	0.073	0.168	0.146	0.056
		RMSE	0.171	0.053	0.122	0.111	0.069	0.193	0.192	0.193	0.194	0.127	0.226	0.227	0.119
	25%	MAE	0.077	0.038	0.053	0.056	0.038	0.074	0.074	0.075	0.075	0.076	0.131	0.105	0.082
		RMSE	0.181	0.056	0.089	0.084	0.057	0.120	0.122	0.122	0.122	0.114	0.167	0.142	0.111
TimesFM	100%	MAE	0.059	0.029	0.055	0.052	0.045	0.094	0.112	0.095	0.097	0.085	0.165	0.148	0.072
		RMSE	0.151	0.043	0.118	0.111	0.080	0.154	0.205	0.153	0.156	0.127	0.203	0.207	0.097
	50%	MAE	0.059	0.025	0.057	0.045	0.042	0.078	0.092	0.077	0.079	0.104	0.177	0.133	0.083
		RMSE	0.153	0.034	0.122	0.094	0.085	0.120	0.157	0.121	0.126	0.150	0.209	0.181	0.119
	33.33%	MAE	0.070	0.033	0.071	0.047	0.043	0.096	0.103	0.098	0.098	0.066	0.151	0.129	0.084
		RMSE	0.177	0.046	0.128	0.096	0.072	0.159	0.176	0.165	0.165	0.118	0.203	0.193	0.112
	25%	MAE	0.075	0.021	0.046	0.053	0.042	0.063	0.063	0.062	0.064	0.083	0.126	0.108	0.078
		RMSE	0.184	0.030	0.090	0.081	0.062	0.110	0.111	0.112	0.114	0.127	0.173	0.161	0.107
MOIRAI	100%	MAE	0.061	0.057	0.059	0.080	0.045	0.079	0.126	0.116	0.122	0.070	0.191	0.097	0.114
		RMSE	0.149	0.073	0.118	0.112	0.075	0.115	0.154	0.143	0.146	0.117	0.229	0.128	0.176
	50%	MAE	0.064	0.026	0.078	0.063	0.039	0.072	0.060	0.071	0.067	0.083	0.182	0.275	0.104
		RMSE	0.148	0.037	0.132	0.098	0.080	0.118	0.096	0.119	0.111	0.128	0.241	0.364	0.149
	33.33%	MAE	0.081	0.012	0.073	0.163	0.046	0.160	0.135	0.145	0.152	0.072	0.180	0.146	0.086
		RMSE	0.173	0.015	0.120	0.229	0.071	0.212	0.177	0.185	0.190	0.119	0.234	0.219	0.132
	25%	MAE	0.078	0.009	0.075	0.084	0.041	0.066	0.066	0.069	0.069	0.104	0.169	0.172	0.085
		RMSE	0.182	0.012	0.100	0.124	0.066	0.103	0.108	0.105	0.107	0.142	0.213	0.247	0.149
Chronos	100%	MAE	0.058	0.029	0.054	0.045	0.044	0.108	0.112	0.106	0.106	0.072	0.119	0.119	0.063
		RMSE	0.152	0.044	0.117	0.088	0.080	0.192	0.202	0.190	0.191	0.113	0.157	0.158	0.106
	50%	MAE	0.055	0.026	0.054	0.041	0.036	0.099	0.116	0.105	0.103	0.068	0.120	0.100	0.072
		RMSE	0.146	0.036	0.118	0.079	0.080	0.179	0.210	0.191	0.188	0.112	0.169	0.139	0.099
	33.33%	MAE	0.068	0.035	0.065	0.041	0.036	0.075	0.081	0.078	0.078	0.063	0.147	0.110	0.094
		RMSE	0.173	0.047	0.125	0.088	0.067	0.134	0.147	0.139	0.140	0.106	0.195	0.165	0.134
	25%	MAE	0.073	0.024	0.047	0.046	0.037	0.106	0.110	0.109	0.109	0.080	0.122	0.104	0.077
		RMSE	0.183	0.034	0.091	0.076	0.059	0.191	0.198	0.199	0.199	0.126	0.166	0.136	0.111
Timer-XL	100%	MAE	0.062	0.038	0.063	0.060	0.034	0.065	0.061	0.063	0.066	0.066	0.148	0.139	0.052
		RMSE	0.148	0.046	0.109	0.106	0.067	0.117	0.112	0.118	0.119	0.120	0.199	0.211	0.109
	50%	MAE	0.061	0.036	0.058	0.051	0.035	0.077	0.079	0.074	0.078	0.075	0.133	0.122	0.069
		RMSE	0.142	0.043	0.107	0.088	0.077	0.109	0.115	0.102	0.110	0.117	0.183	0.172	0.098
	33.33%	MAE	0.070	0.037	0.071	0.048	0.038	0.076	0.076	0.076	0.076	0.066	0.165	0.145	0.054
		RMSE	0.170	0.046	0.114	0.093	0.063	0.120	0.117	0.123	0.121	0.119	0.224	0.218	0.115
	25%	MAE	0.077	0.029	0.048	0.052	0.039	0.082	0.082	0.078	0.081	0.079	0.127	0.110	0.080
		RMSE	0.180	0.035	0.087	0.093	0.058	0.141	0.140	0.135	0.140	0.117	0.167	0.142	0.103

Table A.10
Total training and inference times (in seconds) for all models across the datasets at different sampling rates. For models that were run 10 times, the reported values represent the average training and inference times. For foundation models, only the inference time is reported.

Model	Time (s)			
	25%	33.33%	50%	100%
KNN Regression	8.13	4.65	4.56	4.67
LightGBM	5.29	6.08	16.21	6.12
AutoARIMA	1796.37	13 834.18	30 194.40	37 553.16
Naive Drift	5.36	2.28	2.25	2.41
Naive MA	4.49	4.28	4.64	5.61
NHITS	15.89	18.83	15.00	19.85
NBEATS	15.83	18.57	14.46	19.78
TiDE	12.44	14.86	17.91	24.89
NLinear	10.20	12.36	11.01	16.54
DeepNPTS	11.15	16.50	13.63	27.56
DeepFactor	108.29	121.48	148.33	243.90
DeepAR	220.88	228.49	319.92	545.89
Mamba	347.02	686.86	1089.73	2064.40
TFT	304.35	365.34	501.04	1140.36
MQ-RNN	705.54	1069.40	2215.73	10 045.26
TCN	11.09	12.80	14.37	21.58
BiTCN	13.28	18.56	19.33	33.49
TimesNet	74.15	79.33	150.09	258.18
WaveNet	569.62	720.47	1252.60	2990.42
MQ-CNN	531.44	864.07	1995.80	10 009.17
Informer	244.12	320.31	373.30	753.59
PatchTST	18.44	23.84	32.25	64.83
iTransformer	13.09	18.10	17.00	18.48
Vanilla Transformer	69.36	94.97	133.05	369.01
TimeXer	18.51	21.85	22.74	25.72
TimeGPT	22.29	13.76	14.63	13.45
TimesFM	11.72	10.32	10.53	17.70
MOIRAI	11.83	12.02	13.21	16.19
Chronos	6.46	3.92	4.57	5.09
Timer-XL	11.77	6.30	6.35	6.76

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